



Operator's manual for Attachment



Hydraulic Compressor

Product number: for 300-700 –series loaders, front mount: A35409
for 400-700 –series loaders, rear mount: A35456



Please read this manual carefully before using the equipment, and follow all instructions.

Keep this manual for later reference.

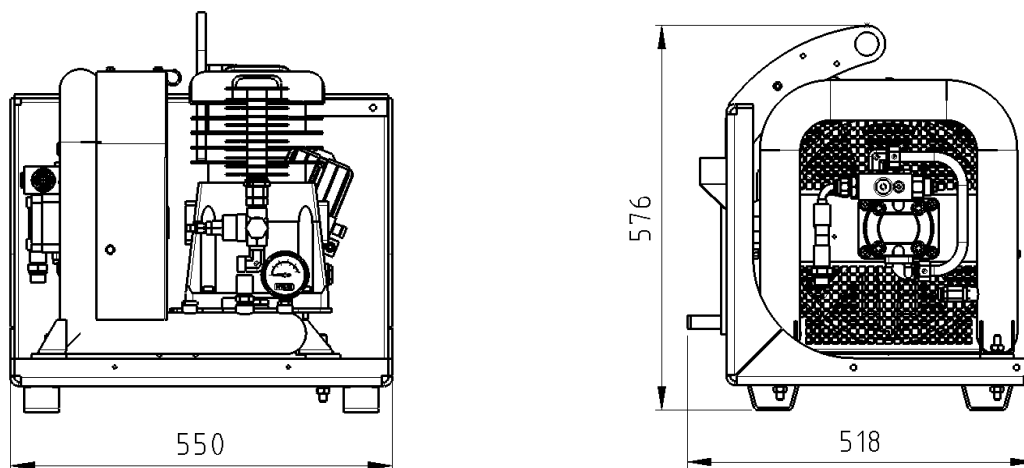
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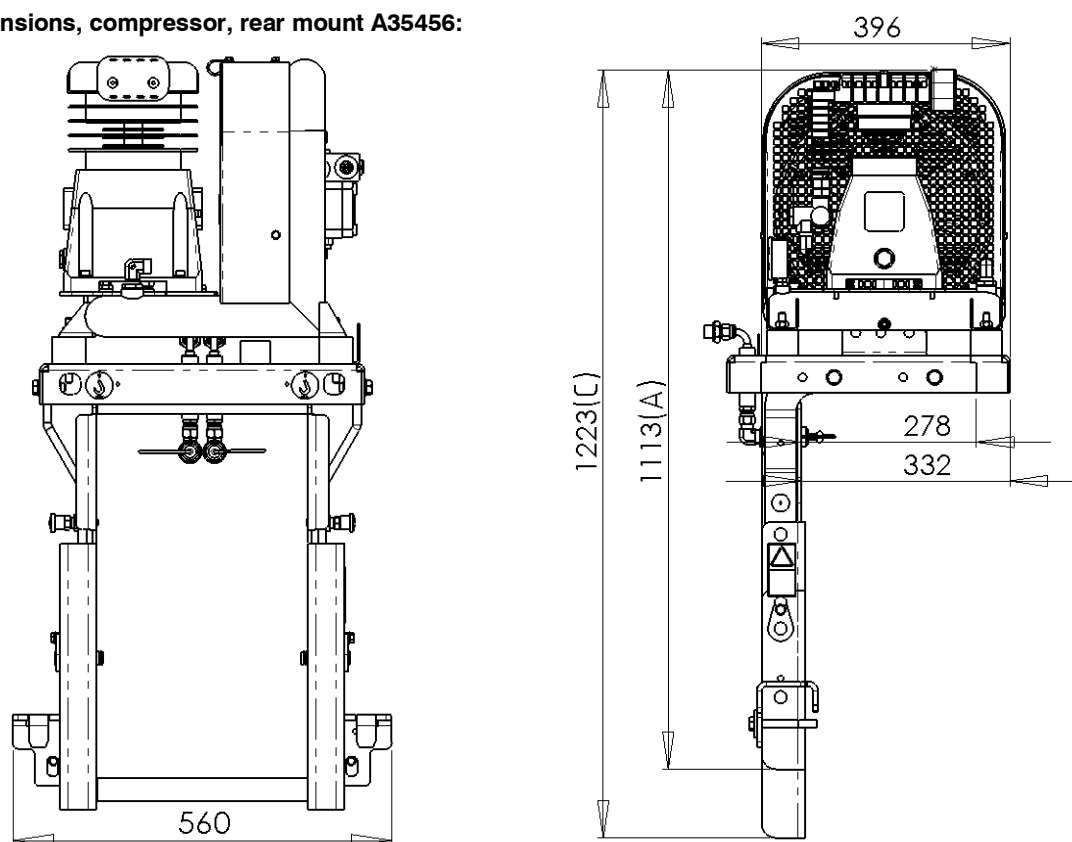
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Dimensions, compressor, front mount A35409:



Dimensions, compressor, rear mount A35456:



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Appendix: Original manual for Dynaset HK400/8-23 compressor unit, including EC Declaration of Conformity



The complete manual for the compressor consists of two parts. The original manual of the compressor unit is supplied with this manual. Make sure that also the original manual of the compressor is also available, cover page shown in adjacent picture.



1. Foreword

Avant Tecno Oy would like to thank you for your purchase of this attachment for your Avant loader. The attachment has been designed and manufactured for easy and productive operation with Avant loaders. By familiarising yourself with this manual and following the instructions, you ascertain your safety and ensure the reliable operation and long service life of the equipment. Read the instructions carefully before starting to use the equipment or performing maintenance. The purpose of this manual is to help you to:

- operate the equipment in a safe and efficient manner
- observe and prevent any hazardous situations
- keep the equipment intact and ensure a long service life

The following warning symbols are used throughout this manual to indicate factors that must be taken into account to reduce the risk of personal injury or damage to property:

	WARNING:
	This safety symbol refers to important safety information in this manual. It warns of an immediate hazard that could cause serious personal injury or property damage. Read the warning text accompanying the symbol carefully and ensure that other operators are also familiar with the warnings, since personal safety is at stake.

NOTICE

This notice symbol indicates information about the correct operation and maintenance of the equipment.

Failure to observe the instructions accompanying the symbol can lead to equipment breakdown or other property damage.

With these instructions, even an inexperienced user can use the attachment and loader safely. The manual includes important instructions for experienced AVANT operators as well. Ensure that all persons using the loader have received proper guidance and familiarised themselves with the manual of the loader, this operator's manual, the original manual for the attachment, and all safety instructions before using the equipment. Using the equipment for other purposes or use in any other way than described in this manual is prohibited. Keep this manual at hand throughout the service life of the equipment. If you sell or transfer the equipment, be sure to hand over this manual to the new owner. If the manual is lost or damaged, you can request a new one from the retailer or manufacturer. Due to continuous product development some details shown in this manual may differ from your equipment.

In addition to the safety instructions included in this manual, you must observe all occupational safety regulations, local laws and other regulations concerning the use of the equipment. Particularly the regulations concerning the use of the equipment on public road areas must be observed. We reserve the right to change the contents of the manual without notification. This operator's manual is original English version.



In addition to this operator's manual, the original manual of the compressor unit must be observed also. In unclear or conflicting points the original manual must be followed.

This annex covers attaching and using the machine particularly with Avant loaders.

2. Designed purpose of use

The hydraulically operated air compressor is intended for use with Avant 300-700 series multi purpose loaders as an easy to use and transportable attachment. Compatible loader models are shown in Table 1 below. The strong and robust compressor makes use of the auxiliary hydraulics oil flow of the loader, making it possible to have high output of compressed air easily near work site even in difficult conditions. The compressor can be used with all typical pneumatic equipment requiring compressed air, such as different hand tools, painting work, construction work, or filling tyres, if the tool or device is compatible with the 8 bar pressure produced by the compressor. If needed, the compressor can be connected to a larger system to provide temporarily compressed as a back-up compressor.

The operating speed of the compressor depends on the loader engine rpm and hydraulic oil output, and in addition, the compressor is equipped with a hydraulic operating speed compensator to reduce the maximum operating speed of the compressor. The compressed air reservoir is integrated to the frame of the compressor and it is equipped with a standard type quick fitting to connect equipment. A pneumatic valve will stop the compressor and set the hydraulic flow to free circulation when the pressure in the reservoir reaches operating pressure. Connecting the compressor block directly to other systems by bypassing the integrated reservoir is prohibited. Depending on the compressor model, it can be attached to the front quick coupling of the loader, or it can be fitted to the back of the loader with mounting bracket, leaving the boom available for another attachment. When using the compressor, it must be attached and locked to the loader. The compressor is intended to produce compressed air, other gas types are not allowed.

The attachments have been designed to require as little maintenance as possible. The operator can perform regular maintenance tasks. However, all repair work can't be performed by the operator, and demanding repair and maintenance operations must be left for professional service technicians. All maintenance work must be done using proper safety equipment. Spare parts must be identical with original specifications, which can be ensured by using only original spare parts.

Familiarise yourself with the manual's instructions regarding service and maintenance. Please contact your AVANT or Dynaset retailer if you have additional questions about the operation or maintenance of the equipment, or if you require spare parts or maintenance services.

Table 1 – Compatibility of compressor models A35409 and A35456 with Avant loaders

Model	216 218 220	313S 320S 320S+	419 420	520 528	525LPG R28 R35	630 635 640	745 750
Front mount A35409	-	•	•	•	•	•	•
Rear mount with bracket A35456	-	-	•	•	-	•	•

When fitting the compressor to the back of the loader, the loader must be equipped with double acting rear auxiliary hydraulics. In this case, the rear side weights can't be used.

3. Safety instructions for using the compressor

Please bear in mind that safety is the result of several factors. The loader-attachment combination is highly powerful, and improper or careless use may cause serious personal injury or property damage. Due to this, you should carefully familiarise yourself with the manuals of both the loader and the attachment before starting operation. Do not use the attachment if you have not familiarised yourself with its operation and the related hazards.



Misuse, careless use, or using an attachment that is in bad condition may cause serious injuries. Familiarise yourself with the controls of the loader and safe coupling and operation of the attachment at a safe area. Become familiar especially with how to stop the equipment in a safe manner.

Read all safety instructions carefully before handling the attachment



- The compressor can be used only when it is attached and locked to the loader. When mounting the compressor to the back of the loader, the rear mounting kit and correctly installed rear auxiliary hydraulics outlet must be used.
- When attaching the attachment to the loader, **ensure that the locking pins of the loader's quick attach plate are positively in the lower position** and that they have **locked the attachment to the loader**. When using the rear mounting bracket, make sure that the bracket is locked to upright position.
- The compressor is to be used by one operator at a time. Prevent others from operating the controls of the loader when compressor is in use. The attachment is intended for adult operators only.
- Ensure that the loader will stay firmly still when leaving the driver's seat, even if the loader is parked on an even surface. The loader could crawl due to vibration or fault in drive circuit resulting in hazard of being run over by moving loader. Always engage the parking brake and use chocks as needed.
- A running loader must not be left unmonitored for long periods. Shut down the loader engine and turn off the auxiliary hydraulics if compressed air is not needed. The temperatures of the loader and attachment must be monitored and overheating must be prevented.
- The exhaust gas of the loader may contain significant amounts of carbon monoxide (CO) and particulate matter (soot), and is hazardous in poorly ventilated areas. The use of diesel or gasoline powered loaders indoors (i.e. garages, storage buildings, construction sites) can be fatal and using such machines indoors is prohibited unless special precautions have been made to remove the exhaust of the loader. Using a loader powered with LPG in poorly ventilated areas can also be fatal because of the possibility of carbon monoxide emissions.
- Never leave the driver's seat when the attachment is lifted high off the ground, to height of over one metre. Always lower the compressor near the ground before leaving the driver's seat. Make sure that the attachment is properly supported during maintenance or inspections. Going under a raised attachment or loader boom is dangerous, as the boom may lower due to loss of stability, mechanical fault or by other person operating the controls of the loader.
- When cleaning, checking or servicing the compressor, the hydraulic pressure and the compressed air must be relieved as their uncontrolled discharge can cause injuries. Follow the safe stopping procedure.



- Running compressor will get hot quickly and touching the surfaces of the compressor can cause burns. Keep clear of hot surfaces and make sure that the cooling surfaces of the compressor are clean. The attachment must be kept clean for efficient cooling and to avoid collecting of combustible dust on the compressor. Keep the compressor away from walls or other obstacles that could have an adverse effect on the operation of the cooling system. Keep the loader and the compressor away from easily ignited materials or dust. Also, make sure that the hydraulic oil cooler of the loader is in good condition and clean. In case the hydraulic oil overheats, the compressor must be stopped and the system must be allowed to cool before restart.
- Never modify the attachment in a way that could affect its safety. The set pressure level must not be exceeded. Only authorised service representative can change the settings of the compressor. Modifying, tampering with, or changing the air pressure safety relief valve to any other type than original spare part may cause risk of bursting of compressed air reservoir and destruction of the compressor unit. The condition and operation of the pressure relief valve must be checked regularly and the valve must be kept clean and in good condition.
- Make sure that you operate only an attachment that is in good condition. For example, damaged compressed air reservoir, bent tubing or wrongly adjusted pressure level together with a faulty pressure relief valve can lead to explosion-like release of compressed air. Never operate the compressor if signs of leaks have been found on the compressor or any pneumatic tool or hose. Do not attempt to repair damaged pressure reservoir, contact service if needed.
- Transport and operate the compressor when the loader boom is lowered to its lowest position and near the loader for the best stability of the loader.
- Operate the controls of the loader only when sitting on the driver's seat. Wrong control movement will cause hazard of crushing to any person nearby the loader outside the driver's area. Always get on the driver's seat to start, stop, or to change the position of the compressor.
- Pay attention to the surroundings and any other persons and machines moving in the vicinity. Pay attention to the contours of the terrain as well as other hazards, such as loose rocks, or branches and trees that reach the driver's area.
- Also read the safety instructions of the loader from the operator's manual of the loader. Read through the original operator's manual of the Dynaset compressor unit as well.



Note that the loader boom moves down and may be tilted even if the engine has been stopped. Never go under a raised attachment or boom or to the danger area between the boom and the attachment. Always lower the attachment near ground before leaving the driver's seat. Lock the parking brake and make sure that the loader will stay still when leaving the driver's seat.



The surfaces of the compressor will get hot quickly. Keep clear especially from the compressor block and keep the attachment clean to ensure that the cooling system works efficiently. Make sure that the air flow around the compressor is not obstructed.



Remember needed safety equipment and to wear proper protective clothing:



- The noise level at the driver's seat and around the machine may exceed 85 dB(A). Wear hearing protection while working with the loader. Also, working with pneumatic tools can produce high levels of noise. To lower noise exposure, keep loader rpm at correct level and use tools that are designed to be as quiet as possible.



- Wear safety boots when working with and around the loader.



- Wear protective gloves.



- When handling hydraulic or pneumatic hoses or components, wear safety goggles. The use of safety goggles is recommended when working with various types of pneumatic tools.



Find out about other necessary safety equipment at specific operating areas. For example, when working at construction sites, safety helmet is required. In dusty conditions wear breathing mask.



Safe stopping of the equipment in case of disruption of operation, before maintenance work, or when preparing the compressor for any storage period



Always stop the compressor following safe stopping procedure before going near the attachment. Never go under a raised loader boom or the attachment. Note that the loader boom can move down even if the engine of the loader is turned off. Safe stopping procedure:

- Stop the attachment and the loader.
- Lower the boom and the attachment on the ground.
- Shut down the loader engine and lock the parking brake.
- Release residual pressure from the hydraulic system.
- Prevent starting of the machine, remove key from ignition.
- Release the compressed air from the reservoir of the compressor (see Chapter 6.4) and relieve air pressure from the hoses or tools connected to the compressor.

4. Technical specifications

Table 2 – Avant hydraulic air compressor specifications

Product number:	A35409	A35456
Weight with mounting frame:	72 kg	66 kg
Compressed air output ISO 1217:	400 l/min	
Operating pressure:	8,0 bar	
Maximum pressure, safety pressure relief valve opening:	9,5 bar	
Type of air connector:	Quick connector BSP R1/2"	
Compressor block type:	Dynaset HK400/8-23 PNE	
Frame reservoir volume:	4,4 l	
Maximum hydraulic pressure:	22,5 MPa (225 bar)	
Nominal hydraulic oil input flow:	25 l/min	
Minimum needed hydraulic oil flow:	12 l/min	
Maximum hydraulic oil flow:	40 l/min	
Rotating speed control	Primary control: loader engine rpm, compressor is equipped with speed compensator	
Hydraulic oil pressure with rated output:	210 bar	
Hydraulic oil pressure when running unloaded:	50 bar	
Operating temperatures:	Standard: -5°C ... 40°C	
Suitable AVANT loader:	See Table 1	

Pressure and compression ratio:

The rated absolute intake pressure is 1 bar (100 kPa). The used reference pressure is barometric pressure at sea level 1,0135 bar, which at an altitude of 100 m from sea level will drop to approximately 0,9 bar. When operating pressure is 8 bar, the compression ratio will be:

$$\frac{8+1,0135}{1,0135}=8,89$$

4.1. Operating conditions

Because of the exhaust emissions of any type of loader, the use of compressor is allowed only outdoors or in areas where special measures have been taken to ventilate exhaust gases. The compressor must not be used at areas where there is smoke, toxic or flammable fumes, or other impurities in the air.

Permissible operating temperature:

With factory oil, the compressor can be used in -5°C ... 40°C outdoor temperatures. The compressed air produced by the compressor is 5 to 10°C warmer than surrounding air. When temperature drops below 0°C , the compressed air reservoir must be emptied after each use (see Chapter 6).

When alternative type of compressor oil is used, shown in Chapter 7, the compressor can be used in conditions according to oil type. According to standard EN 1012-1 the maximum allowed compressor oil temperature is 100°C . In normal operating conditions with 20°C outdoor temperature, the normal compressor oil temperature is around 80 - 85°C .

Humidity:

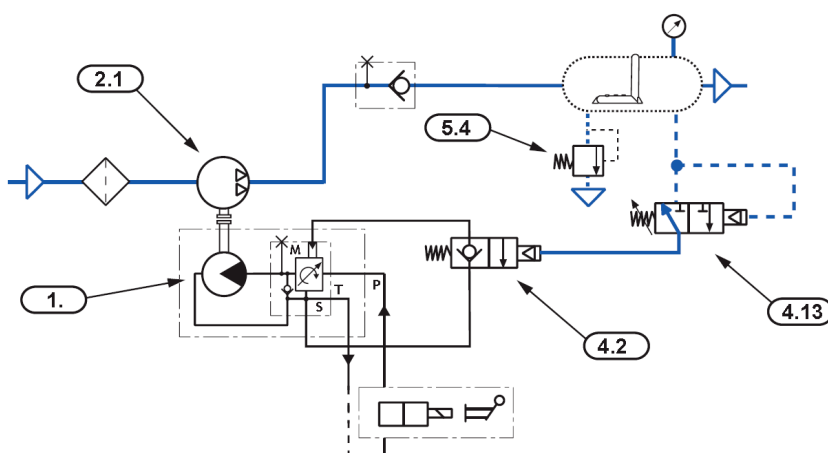
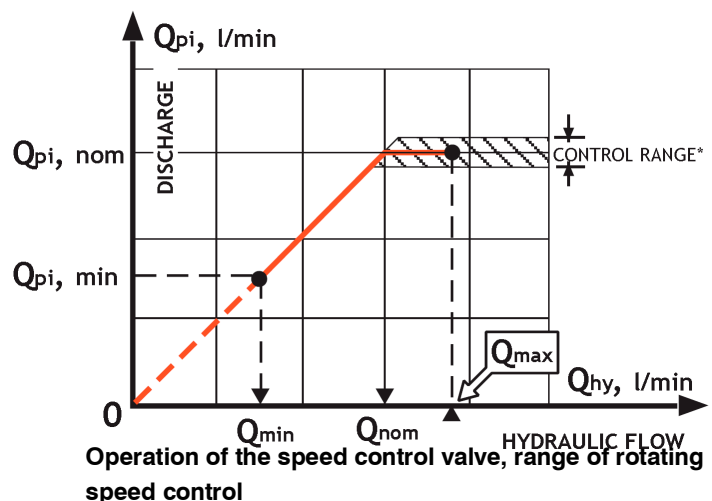
The compressor can be operated with maximum air humidity of 90%, while temperature is within permissible range. If humidity exceeds 90%, contact service.

4.2. Speed control system and automatic stopping (PNE)

Speed control system

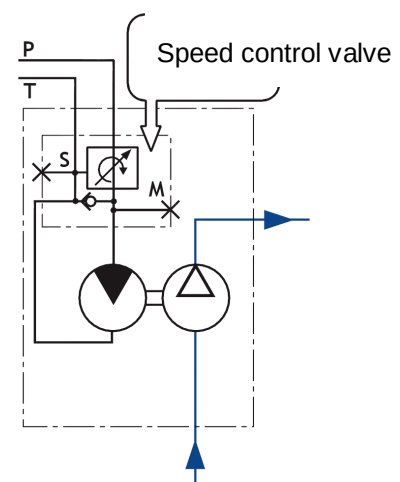
The speed control valve, integrated to the hydraulic system of the compressor, regulates the rotating speed of the compressor so that the maximum input can be exceeded by 10 to 20 l/min as shown in the figure on the left. This will help to achieve the best performance of the compressor.

The speed control valve is automatically and individually functioning component. However, the nominal rated input should not be exceeded, and it will not increase the performance of the compressor, but will unnecessarily heat up the hydraulic oil. The system includes a valve that prevents the compressor from rotating in the wrong direction.



Main Hydraulic and pneumatic components of the compressor

See detailed descriptions in Dynaset manual

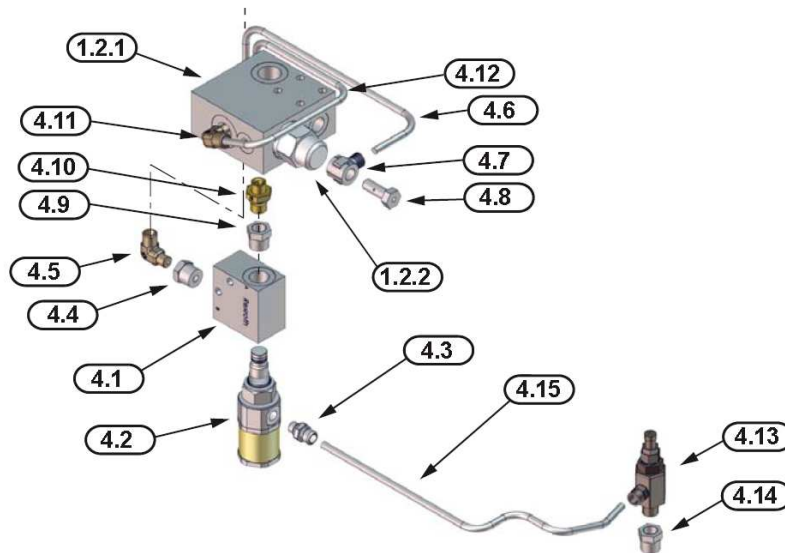


Connection of speed control valve within the hydraulic system

Automatic stopping system

The air pressure in the compressor tank controls the integrated automatic stopping system of the compressor. When functioning normally, the air pressure gauge will reach set pressure level, and the compressor will stop. When the stopping system stops the compressor, the air pressure between the compressor block and tank is vented, making it easier for the compressor to restart. Once the pressure in tank has dropped about 2-4 bar, the compressor will restart smoothly, delivering uninterrupted flow of pressurized air.

The system is pre-adjusted and does not include components that are serviceable or adjustable to user. Check regularly that the components are clean and in good condition. Make sure that air hose shown as 4.15 in figure below is in good condition, to ensure reliable operation of the system. The stopping system sets the hydraulic circuit to free circulation with valve 4.2, when the pre-set air pressure level is reached in valve 4.13. More detailed component descriptions can be found from the operator's manual of the compressor unit.



Components of speed control valve and automatic stopping system

4.3. Safety relief valve

The safety relief valve 5.4., shown in figure on previous page, must be an original part in good condition. Check the condition of the safety relief valve visually for dirt, ice, corrosion, or other possible changes that may reduce the service life of the valve or obstruct the valve from functioning normally.

The opening of the safety relief valve can be observed as loud noise related to quick release of air, and if the pressure gauge shows higher pressure than nominal operating pressure. If the safety relief valve is active, or the pressure reaches higher than operating pressure, stop the use of the compressor immediately and investigate the cause of high pressure. Contact service. The opening pressure of the safety relief valve must never be changed.

4.4. Warning labels

Listed below are the labels and markings, which must be visible on the equipment. Replace the warning labels, if they have become unclear or if they have detached completely. New labels are available via your retailer or contact information provided on the cover of this manual.



The warning labels contain important safety information and they help to identify and remember the hazards related to the equipment. Replace any damaged or missing warning label with new one.



A46771

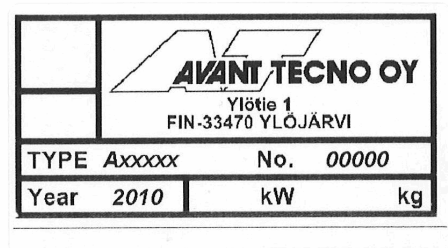


A46772



A46803

Only when the compressor is installed to the rear of the loader.



Identification plate of the attachment

4.5. Main components of the compressor

Table 3 – Compressor, front mount

Main components A35049

1	Power Pack front mount frame A35406
2	Compressed air connector
3	Reservoir pressure gauge
4	Frame air reservoir
5	Automatic speed control valve
6	Air filter
7	Oil level sight glass
8	Oil fill port
9	Reservoir drain, manual valve

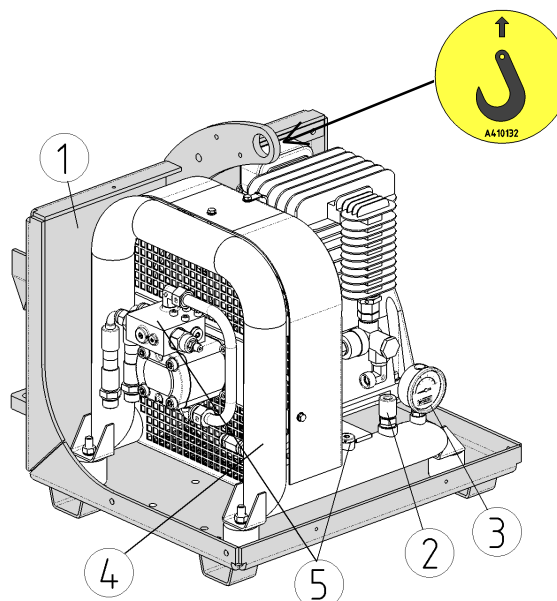


Table 4 – Warning labels and locations, front mounted compressor

Decal	Warning message
A46771	Read instructions before use.
A46772	Do not go under a raised attachment or loader boom; stay away from raised equipment.
	Attachment identification plate

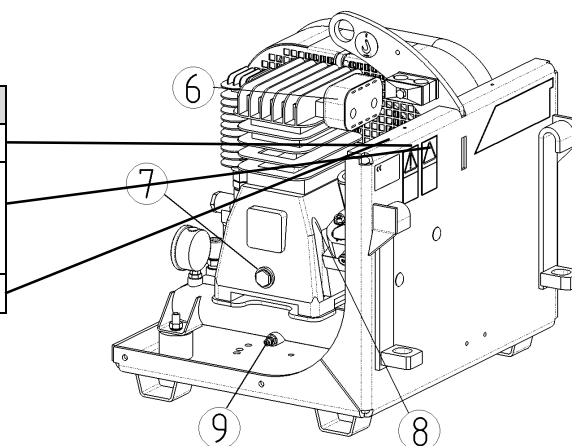


Table 5 – Compressor, rear mount frame

Main components A35456

10	Power Pack rear mount frame A35546
11	Frame tilt locking
12	Hydraulic hose stand
13	Rear mount frame fitting brackets
14	Rear mount frame screws

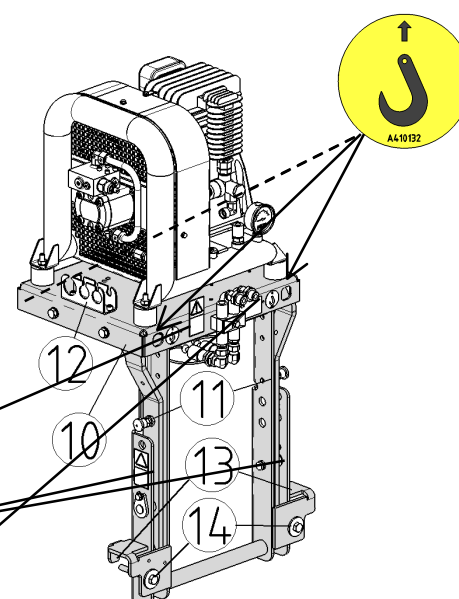
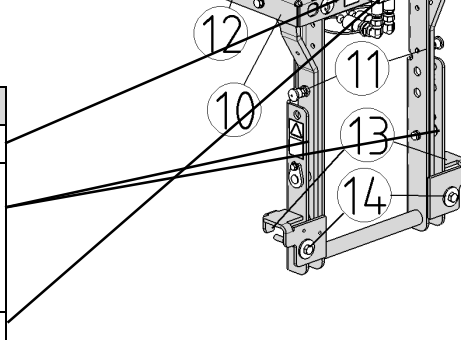


Table 6 - Warning labels and locations, front mounted compressor

Decal	Warning message
A46771	Read instructions before use.
A46803	Crushing hazard, moving and locking frame mechanism. Keep fingers away from between frame components when tilting the mounting frame.
	Attachment identification plate

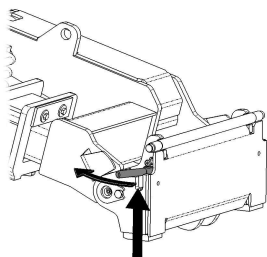


5. Assembly of compressor

The attachment is ready for use as delivered. The rear mount frame can be installed as shown in Chapter 5.4 to a loader that isn't fitted with rear side counterweights or rear lift device.

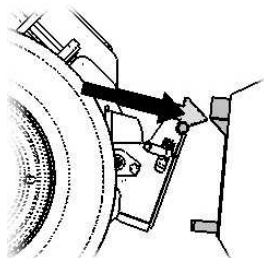
5.1. Mounting the compressor to the loader boom

Mounting the attachment to the loader boom is quick and easy, but it must be done carefully. An attachment that has not been locked may come off from the loader and cause hazardous situations. The attachment is mounted to the loader boom by using the quick attach plate and the counterpart on the attachment. In the worst case, the attachment may slide onto the driver along a raised boom. For this reason, the attachment must under no circumstances be raised to a height of over one metre when it has not been locked. To prevent hazardous situations, always follow the attachment mounting instructions provided in the following pages. Also remember the safety instructions in Section 3 and the instructions provided in the operator's manual of the loader.



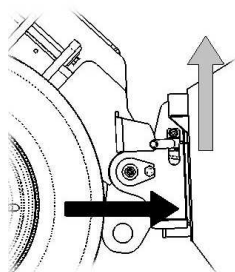
Step 1:

- Lift the quick attach plate locking pins up and turn them so that they are locked in the upper position.
- If your loader is equipped with a hydraulic attachment locking system, see additional instructions about the use of the locking system from the relevant manuals.
- Ensure that the hydraulic hoses are not in the way during installation.



Step 2:

- Turn the quick attach plate hydraulically to an obliquely forward position.
- Drive the loader onto the attachment. If your loader is equipped with a telescopic boom, you can utilise this.

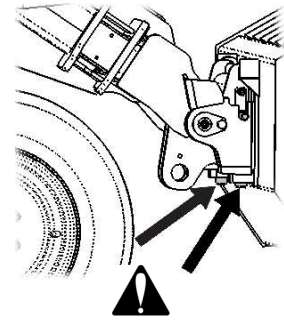


Step 3:

- Turn the boom control lever left to turn the bottom section of the quick attach plate onto the attachment.
- Lift the boom slightly – pull the boom control lever backward to raise the attachment slightly off the ground.
- Turn the boom control lever more to the left to engage locking pins, or lock the hydraulic locking.
- Always make sure that the attachment is fully locked.



An attachment that has not been completely locked to the loader may fall on the boom or operator, or under the loader while driving, causing serious injuries. Before moving or lifting the attachment, make sure that the locking pins are in the lower position and come through the fasteners on both sides. Never move or lift an attachment that has not been locked.



Excessive tilting or lifting of an unlocked attachment increases the risk of tipping the attachment over. Do not use the automatic locking of the locking pins when the attachment is lifted more than one metre from the ground. If the locking pins do not return to the normal position when tilting, do not tilt or raise the attachment any more. Lower the attachment on the ground and secure the locking manually.

5.2. Connecting and disconnecting hydraulic fittings

The default hydraulic connector for front mount compressor is the multiconnector system. If you want to change the type of the hydraulic couplings of the attachment or the loader, contact your Avant dealer or service point for instructions or installation services.



Never connect or disconnect quick couplings or other hydraulic components while the system is under pressure or if the control lever of the auxiliary hydraulics control lever is locked on. Connecting or disconnecting the hydraulic couplings while the system is pressurized may lead to unintended movements of the attachment, or ejection of high-pressure fluid, which can cause serious burns or other injuries.

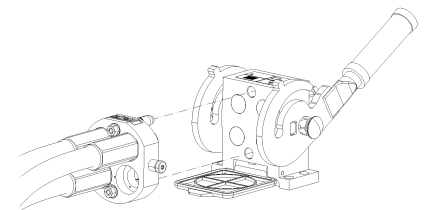
Keep the fittings as clean as possible; use the protective caps for the quick couplings on both the attachment and the loader. Dirt, ice, etc. may make using the fittings significantly more difficult. Never leave the hoses hanging on the ground; place them on top of the equipment.

Connecting the multiconnector system:

Align the pins of the attachment connector with corresponding holes of the loader connector. The multiconnector will not connect if the attachment connector is upside down. Connect and lock the multiconnector by turning the lever towards the loader.

The lever should move easily all the way to its locking position. If the lever does not slide smoothly, check the alignment and position of the connector and clean the connectors. Also shut down the loader and release the residual hydraulic pressure.

To disconnect the multiconnector system, turn off the auxiliary hydraulics of the loader and unlock the multiconnector. To unlock, push the knob on the side of the connector, and turn the lever to disconnect.



Align the pins of the multiconnector system

Conventional quick couplings:

- Used always when fitting the compressor behind the loader.
- Alternative couplings for some loader models.

Before connecting or disconnecting the standard quick couplings, the residual pressure must be released as shown below. The conventional quick couplings will not connect, if there is pressure in the hydraulic system.

To connect and disconnect the standard couplings, move the collar at the end of the female fitting. The hoses should be connected so that the fitting equipped with a coloured cap is connected to the corresponding fitting of the loader. Note that the protective caps on the loader and the attachment can be fastened to each other during operation to reduce the accumulation of dirt. When disconnecting the standard quick couplings a small amount of oil may drip from the couplings. Wear protective gloves and have some cloth at hand to keep the equipment clean.



Conventional quick couplings

NOTICE

Make sure that both quick coupling fittings are properly connected, otherwise the hydraulic motor of the attachment may get damaged.

Releasing residual hydraulic pressure:

To make sure that there will not be residual pressure in the hydraulic system of the attachment, shut down the loader engine and move the auxiliary hydraulics control lever of the loader to it's extreme positions before disconnecting the couplings. If the fittings will not connect to the loader, the residual pressure must be released. The residual pressure must be released before handling the rotating components of the attachment and before storage of the attachment.

5.3. Uncoupling and storage of the compressor

The compressor must be stored in upright position. Avoid storing the attachment directly against the ground, place a pallet or blocks of wood under it, for example. Protect the compressor from rain. When storing the compressor, empty the air reservoir and drain all condensed water from it as shown in Chapter 6.4. Also, read instructions concerning maintenance and service from Chapter 7.

NOTICE

When uncoupling the attachment from the loader, always disconnect the hydraulic fittings first before unlocking the quick attach plate, to prevent hose damage and any oil spills. Reinstall the protective caps on the fittings to prevent dirt from entering the hydraulic system.

NOTICE

Avoid storing the attachment directly on the ground for long periods to avoid rust damage. Protect compressor from rain. Service the compressor as shown in chapter 7.

5.4. Installing the rear mount frame

The compressor can be installed to the back of 400-700 series loaders with the rear mount frame delivered with compressor A35456. The loader must be equipped with double acting rear auxiliary hydraulics to connect the compressor. The rear mount frame is fitted to the rear counterweight or light rear bumper. Extra counterweights can't be used and also the trailer coupling must be removed.

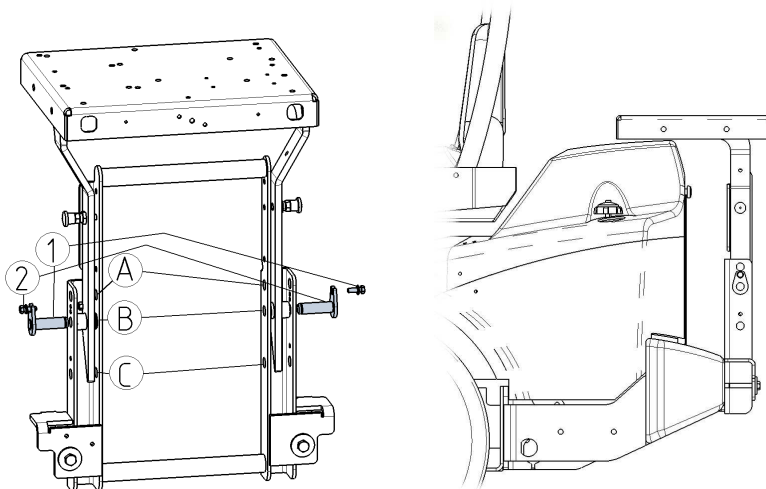
To install the rear mount frame, a suitable lifting device or assisting person is needed, the compressor with mounting bracket weighs about 66 kg. Install the compressor on flat ground and allow the loader to cool down first.

1. Check the adjustments of the frame to match the loader.
2. Remove possible extra counterweights from the loader.
3. Clear the threads on the rear counterweight or light bumper in advance.
4. Lift the frame on the counterweight or light bumper.
5. Tighten fitting bolts and make sure the frame is locked upright.
6. Connect hydraulic hoses to rear outlet on the loader. Make sure that the hoses will not abrade against sharp corners, and make sure that the hoses are not close to the exhaust of the loader or other hot points.

Height adjustment of the rear mount frame:

The frame is adjusted by removing the pin locking screws (1) and the pins (2) shown in the adjacent figure. Before installing the frame to the loader, make sure that the bracket is adjusted as follows:

Frame setting	Loader models
A	419 420 520
B	528 630 635 640
C	745 750



Check also the longitudinal adjustment of the frame. The frame can be adjusted closer to the loader only for 700 series loaders. In other loader models the frame could hit the cabin or seat frame, if not in the rear setting.



Make sure that the rear mounting frame is locked on the loader. Falling frame creates hazards of crushing, oil spill and burns. Also, ensure that the frame is locked on both sides of the frame whenever the loader is in use.

6. Instructions for use

When the compressor and its hydraulic hoses are connected to the loader as shown in his manual, the compressor is ready after initial checks.

When the compressor is needed, leaving the driver's seat is allowed only when the loader is parked on firm and level ground so that the moving of the loader is prevented by using suitable means in addition the parking brake as needed. The loader boom must always be lowered. Other persons must not be allowed to reach the driver's seat of a loader that has been left running to operate the compressor, so that the loader will remain still in safe position while in use. The loader must not be left running for long periods. Things to monitor include the hydraulic oil and coolant temperatures of the loader.



Unintended moving of the loader or wrong control movement of the controls cause hazard of crushing. Start or stop the attachment only when sitting on the driver's seat, when the compressor is in upright position close to ground surface. Always lock the parking brake, turn the articulated frame straight and ensure that the loader will stay still and stable before leaving the driver's seat. Prevent other people from reaching the controls of the loader.



6.1. Checks before use

- Operate the compressor only when the loader is parked and the boom is lowered.
- Ensure that during use the compressor is in upright position.
- Check the condition of the compressor, leakages, cleanliness, cooling fan, and oil level. See list of checks in Chapter 7.2 and other instructions regarding maintenance, service and checks in chapter 7.
- Only use tools that are in good condition and make sure that the connected pneumatic systems are safe. Ensure that the connected tool or system will work with the 8 bar pressure produced by the compressor.

6.2. Starting the compressor

Start the compressor by turning the auxiliary hydraulics control lever toward its locking position. There are no adjustable settings on the compressor. Correct operating speed is adjusted by controlling the loader engine rpm, to set auxiliary hydraulics output of the loader to correct level (23-25 l/min, see operator's manual of the loader).

The speed control valve will compensate excess hydraulic oil flow of about 10-20 l/min. However, the maximum input must not be exceeded and the output flow of the loader must be set as correct as possible. If the loader is equipped with two auxiliary hydraulic pumps, using both pumps is possible only when the loader engine rpm is low.

NOTICE

Unnecessarily high engine rpm will heat the hydraulic oil. Set the engine rpm and auxiliary hydraulics oil output to correct level for compressor.

NOTICE

Start the compressor only when it's in upright position to ensure proper oil circulation within the compressor. If the compressor has been tilted considerably, allow it to stand upright for a few minutes before starting it. Avoid tilting the compressor by more than 45 degrees in all situations.

NOTICE

The hydraulic oil cooler of the loader must be kept clean. When working with a 300 series loader without an oil cooler, the compressor can be used only for limited periods at a time.

During use:

Monitor operation of the compressor and loader during use. Use compressed air as instructed in the manual of the specific pneumatic tool and use correctly sized hoses to minimise pressure losses. Never direct air jet towards persons or try to hold leakages with hands; pneumatic air can penetrate skin causing life threatening condition. The compressed air produced by the compressor must not be used as breathing air in any application. Blowing air towards clothes causes some materials to burn much more easily, using blowgun to dry clothes is prohibited.



Do not leave the compressor running unattended. Stability and temperature must be monitored during use.



Indoors or in poorly ventilated areas the exhaust fumes can reach toxic levels and lead to suffocation. Operating the loader indoors is prohibited.

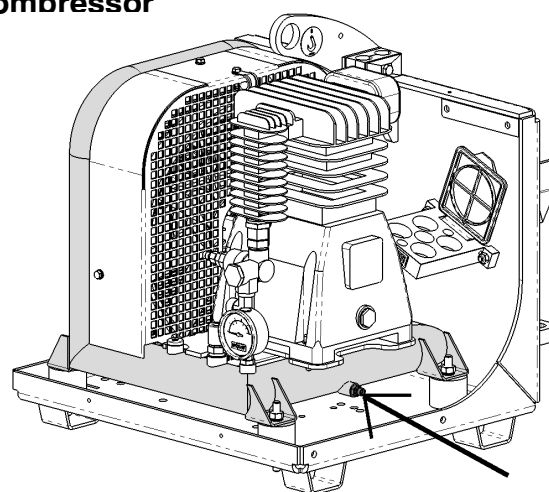
6.3. Stopping the compressor

Stop the compressor whenever compressed air is not needed. The compressor must be stopped by turning the auxiliary hydraulics control lever to its neutral position, or by turning off the loader, if needed. The compressor must never be turned off by uncoupling its hydraulic hoses. The residual air reservoir can be emptied as shown in chapter 6.4. It is recommended to release pressure from any hose or tool to prevent their accidental use. Follow instructions shown in page 8 concerning the safe stopping of the compressor.

6.4. Draining air reservoir and storage of compressor

The air reservoir can be drained by opening the manual valve pointed in adjacent figure. Open the screw slowly and gradually with fingers so that air is released in controlled way. When air has been drained, open the valve more and, if needed, tilt the compressor to drain all condensed water from the reservoir. Tighten the valve using fingers. Do not use tools. Wearing gloves is recommended.

Especially in humid environment or low temperatures the reservoir must be drained from condensed water after each use. The reservoir must be drained before storage.



Manual compressed air tank drain valve

7. Inspections, maintenance and service

The attachment has been designed to be as maintenance free as possible. Continuous consists of regular cleaning, checking oil level and keeping the air filter in good condition. In addition, the condition of any hydraulic or pneumatic component must be monitored.



Ensure that the attachment is properly supported even when it is attached to loader. All maintenance work must be done after the attachment is lowered down completely or when rear mount frame is locked upright. The loader boom can lower unexpectedly causing crushing hazard. Release the residual hydraulic and pneumatic pressures before maintenance.

NOTICE

The compressor must be kept clean for safety and long and reliable operation. When performing maintenance or removing components it must be made sure that dirt will not get into the pneumatic or hydraulic circuits.

7.1. Checking hydraulic and pneumatic components

Check the condition of the hydraulic hoses and components when the engine has been turned off and the residual pressure has been relieved. Repair all leaks immediately after detecting them; a small leak can quickly grow into a big one. Do not use the equipment, if you have discovered a leak in the hydraulic system of the attachment or the loader. Leaking hydraulic fluid may penetrate skin and cause serious injuries. Seek medical attention immediately in case hydraulic fluid penetrates the skin. Wash any part of body that has been in contact with hydraulic oil carefully with water and soap. Hydraulic fluid is also harmful to the environment and any leak to the environment must be prevented. Operate the attachment only with type of hydraulic oil that is accepted for use in Avant loaders.



Never handle hydraulic components when the hydraulic system is pressurized, since a fitting may break or become loose and the released oil may cause serious injuries. Do not use the equipment, if you have discovered a fault in the hydraulic system.



Check hoses visually for cracks or abrasions. Monitor the wearing of the hoses and stop the use, if the surface layer of any hose has worn off. Check the routing of the hoses. Adjust the hose clamps to avoid abrasion to the hoses. Hydraulic hoses have limited service life, which is marked on the hoses. After the date marked on the hoses, they should be replaced with new ones. If there are signs of oil leaks, to check a component, hold up a piece of cardboard in the area where a leak is suspected.

Finding any of these faults means that the hydraulic hose or component must be replaced. Spare parts are available from your nearest AVANT retailer or service point. Leave the repair work to professional service technicians, if you don't have adequate knowledge and experience about hydraulic assemblies and how to perform the repairing safely.

7.2. Checking the compressor and regular maintenance

The condition of the compressor must be checked visually for defects. If any of the following faults is observed on the compressor, discontinue use until the fault is corrected:

- Leaking hydraulic system.
- Damaged quick connector or tool.
- Damaged reservoir pressure gauge.
- Improperly functioning air filter.
- Ambient temperature outside specified range.
- Compressor cooling restricted or compressor overheats (oil temperature >100°C).
- Frozen condensed water in reservoir
- Pneumatic stopping system allows the reservoir pressure to reach safety relief valve opening pressure.
- Safety relief valve modified, its operation has been restricted or setting changed from original.
- Operating pressure higher than rated operating pressure.
- Compressor noise is abnormal.
- Compressor will not produce compressed air, faulty operation.
- Frame reservoir bent, damaged or rusty.

The operator can perform normal checks, basic maintenance and component replacements. However, when working with hydraulic components, qualification requirements must be met.

Qualification requirements for handling hydraulic components:

Replacing hydraulic components must be left for professional service technicians. When handling hydraulic components, special attention to safety is needed, and installation work is allowed only for qualified persons. The speed control valve system of the compressor does not include user serviceable components. The manufacturer or its representative is not responsible for damages, losses or injuries resulted from incorrect or incomplete installations. Contact your nearest Avant dealer or service point, if you have any additional questions or require repair services.

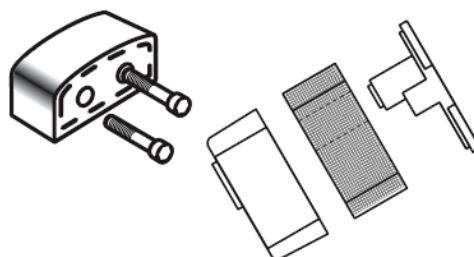
Checking automatic stopping system and safety relief valve:

Monitor the operation of the automatic stopping system and check from the pressure gauge that the compressor will stop when the set 8 bar pressure level is reached. Check the tightness of the air hose 4.15 regularly. See further instructions in chapters 4.2 and 4.3.

7.2.1. Compressor air filter

Clean the air filter at least after every 40 hours of use. Clean the surface surrounding the filter carefully and watch out that dirt will not enter the compressor when removing air filter.

In normal operating conditions the air filter must be replaced after every 500 hours of use, or every three months in frequent use. In demanding or dusty conditions the service interval must be shortened.



Air filter assembly of the compressor

7.3. Cleaning

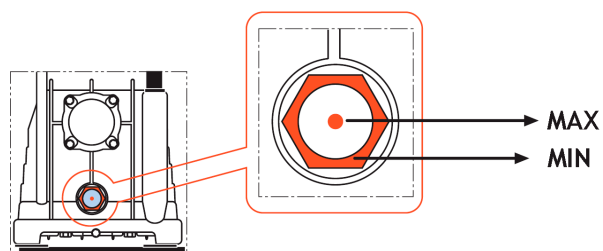
Keep the compressor clean to ensure effective cooling and for the best performance of the compressor. A blowgun is a useful tool to clear dust from all surfaces of compressor. Especially the compressor block, control valves, safety relief valve, air filter, and cooling fan must be kept clean. Cleaning with water should be avoided. The compressor must not be washed with a pressure washer. Touch up paint where needed to prevent rust damage.

7.4. Oils

Checking oil level

To check the oil level there is a sight glass on the side of the compressor block. For correct reading, the level must be checked in upright position after the compressor has been stopped for long enough to allow oil to drain back to oil pan.

The oil level must be between the lower edge and the middle of the sight glass.



Oil level sight glass markers

Oil type	Viscosity	Operating temperature	Change interval, maximum	Oil volume
Multi-grade	SAE 10W-30	-5°C ... +40°C	500 h / 12 m *	0,91 l
Diesel engine oil API CC MIL-L-2104 MIL-L-46152 B	SAE 10W	-25°C ... +20°C	500 h / 3 m *	
	SAE 30	-5°C ... +30°C	500 h / 3 m *	
	SAE 40	+5°C ... +40°C	500 h / 3 m *	

*The first oil change must be made after a maximum of 150 operating hours.

7.5. Removal from service

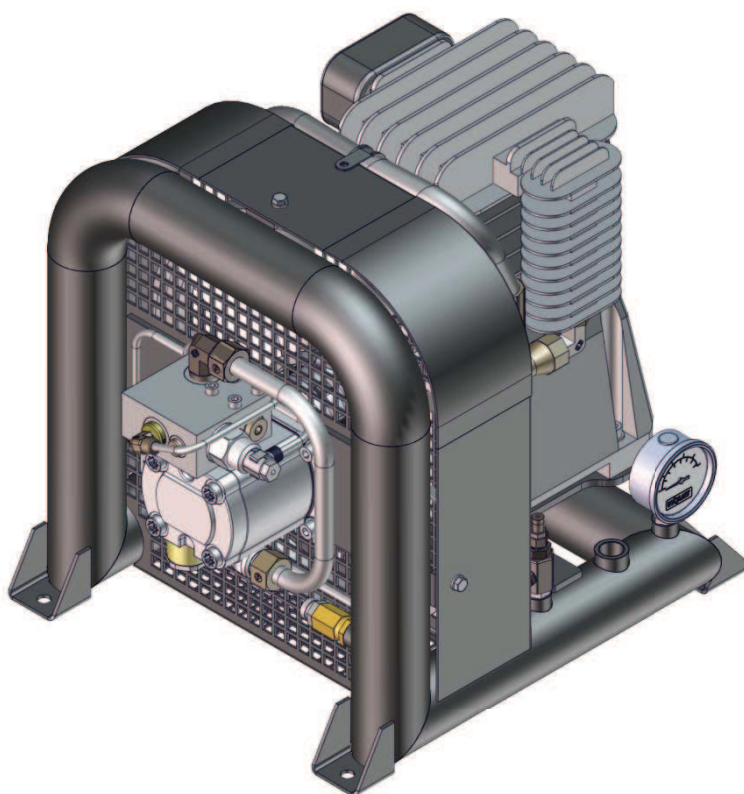
Follow local laws, guides and recommendations concerning waste disposal and recycling of materials. Before disposal of the compressor, remove all oil and filters and dispose of them properly. Separate compressor parts according to their material and give them to proper collecting facilities. Customers can send used equipment to Avant Tecno Oy for disposal or recycling at their own cost. All sent material must bear customer contact information.

7.6. Troubleshooting

1	Compressor will not run, even when loader is running with auxiliary hydraulics turned on	1.1	Badly or wrongly connected hydraulic hoses, or fault in loader valve	Check hose connectors and operation of hydraulic valve.
		1.2	Excessive compressed air pressure	Release pressure by using a pneumatic tool or with manual drain valve.
		1.3	Fault in hydraulic system of compressor	Make sure that oil pressure and flow input are within specifications. Make sure that there are no leaks. Contact service.
2	Compressor starts hardly	2.1	Excessive compressed air pressure	Release pressure by using a pneumatic tool or by opening manual drain valve.
				Check stopping system components.
3	Air pressure does not reach set level	3.1	Faulty safety valve or wrong setting	Check condition and setting of safety valve. Replace if necessary.
		3.2	Fault in hydraulic system	Make sure that oil pressure and flow input are within specifications. Adjust as needed. Check the condition of hydraulic motor and possibility of internal leaks. Contact service to check or repair hydraulic motor.
		3.3	Clogged air filter or intake manifold	Check, clean and/or replace.
4	Compressor does not create pressure	4.1.	Air connector damaged or tool failure	Check and fix possible fault.
		4.2	Speed control damaged or adjusted incorrectly	Contact service.
		4.3	Clogged air filter or intake manifold	Check, clean and/or replace.
5	Excessive oil consumption	5.1	Wrong type of oil	Use oil suitable for operating conditions.
		5.2	Worn piston rings	Check and fix possible fault.
6	Compressor overheats	6.1	Dirty compressor block, clogged cooling fan grill	Check and clean.
		6.2	High ambient temperature	Extra cooling must be arranged for compressor.
		6.3	Low oil level	Add oil to specified marker.
		6.4	Overheating hydraulic oil heats compressor	Allow hydraulic system to cool before continuing. Check operation and condition of loader oil cooler.

HYDRAULIC COMPRESSOR

HK400/8-23_PNE



OPERATING INSTRUCTIONS

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SAFETY PRECAUTIONS



The pressure in both hydraulic and compressed air circuits of HK-compressor is considerably high. There at the technical condition of your equipment should be under constant scrutiny. Operators and maintenance personnel must always comply with local safety regulations and precautions in order to close out the possibility of damages and accidents.

Especially couplings, valves and hoses should be maintained tight and clean as well as kept under constant observation avoid possible leakages of pressure and hot blowouts. Hydraulic leakages must be rectified immediately to avoid user and bystander injuries. Follow all your local safety instructions related to the high pressure hydraulics.

In order to exclude possible accidents, it is not allowed to clean or inspect HK-compressor and pneumatic tools when hydraulic or/and pneumatic circuit is pressurized. Prior to any cleaning, inspection and service hydraulic system of your carrier machine must be stopped and both hydraulic and pneumatic circuits dissipated, for instance, by opening the blow gun.

Always wear appropriate clothing and safety equipment such as goggles, ear protection and safety shoes at all times when operating the compressor. When operating the compressor, beware of machinery parts warmed by hot hydraulic oil.

EXTERME CLEANLINESS MUST BE MAINTAINED WHEN CARRYING OUT ANY SERVICE REPAIR DISSASSEMBLING OF HK-COMPRESSOR AND HYDRAULIC SYSTEM. THIS IS CRUCIAL TO ENSURE SAFE, RELIABLE AND LONG-LIFE OPERATION OF YOUR EQUIPMENT.

Hydraulic system of a carrier machine should be maintained according to the service program. All installation and service of both hydraulic and electric equipment must be performed by qualified and experienced personnel only.

WARNING STICKER

Dynaset dispatch department includes one (1) warning sticker bag per one (1) main product. Product recipient is obligated to fix determinate warning sticker to Dynaset product. Fix sticker to visible and appropriate place or close to Dynaset product where it's easily seen. Before fixing sticker clean surface with solvent detergent.



READ OPERATION INSTRUCTIONS.



GENERAL CAUTION.
BEWARE OF PRESSURE AIR.



BEWARE OF HOT SURFACE. USE EAR PROTECTOR.



GENERAL NOTES

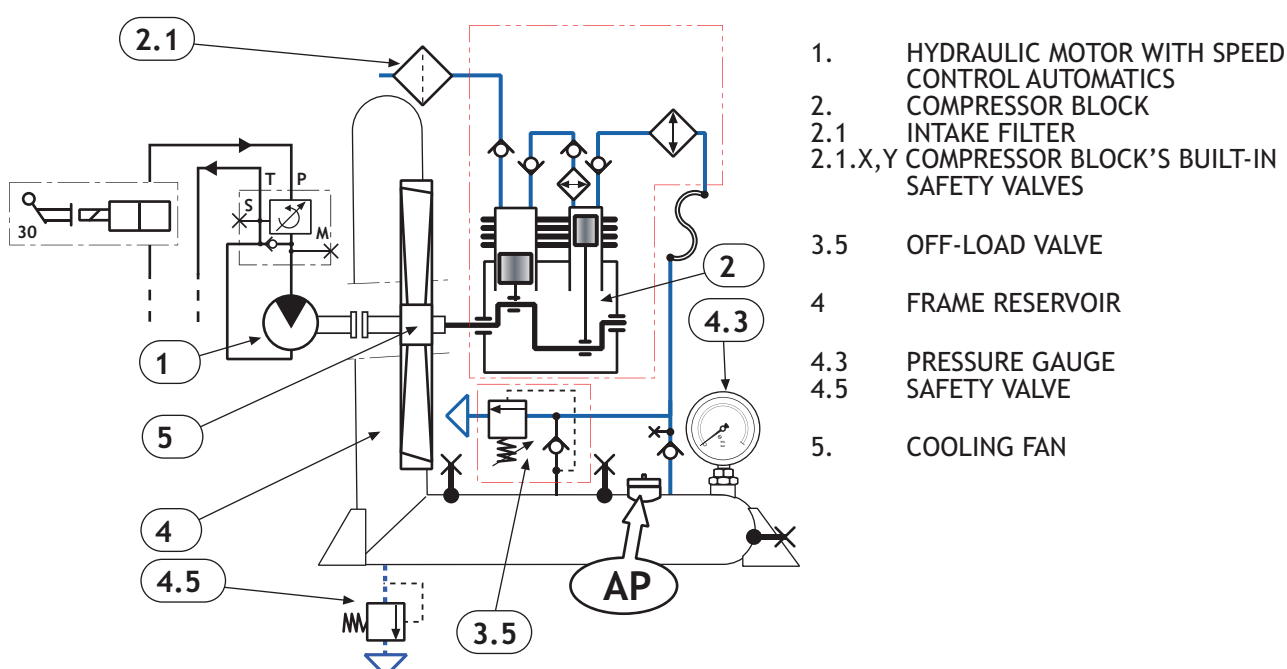
DYNASET hydraulic compressors are compact and integrated all-in-one units, especially designed for mobile installation. The only power source is a hydraulic system to provide compressor with required hydraulic fluid flow at demanded pressure.

DYNASET hydraulic compressor transforms hydraulic power into a high quality compressed air.

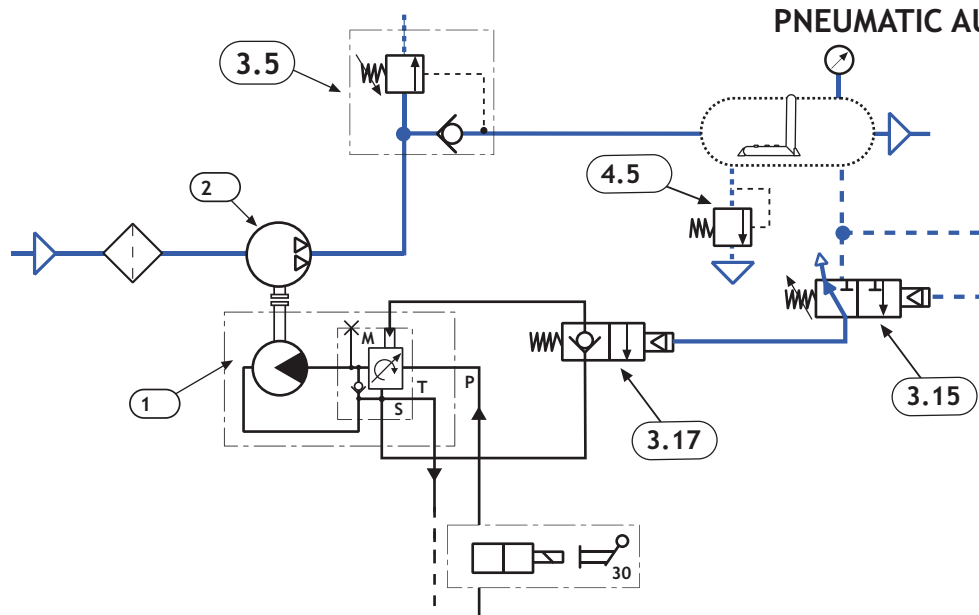
Designed to meet regular compressed air demands, HK hydraulic compressors are used to power pneumatic tools, for cleaning, flushing etc.

CONSTRUCTION

DYNASET hydraulic compressor of HK-series is mobile unit comprised of a hydraulic motor with incorporated automatic rotation speed control valve (except -S models) and 2-piston compressor block. Compressor block may be either of single or double stage construction depending on the required output characteristics. Compressor's basic assembly includes also frame reservoir, unloading pressure regulator, safety valve and pressure gauge.



PNEUMATIC AUTO-OFF -PNE



FUNCTION

While the compressor's crankshaft is rotated by a hydraulic motor, air taken through a filter into a compression chamber of two-cylinder block - either single or double stage construction - is compressed and delivered through aftercooler and frame reservoir further to the consumer.

The fan, mounted on a motor shaft, provides compressor block as well as both inter- and aftercooler with a proper cooling.

Compressor's frame reservoir is equipped with an unloader valve, safety valve and pressure gauge.

If the compressed air is not used while compressor is running, air pressure in frame reservoir achieves its maximum when the unloader valve lets off the air (standard assembly) to the atmosphere.

Non-return valve prevents the compressed air, already contained in the system, to return into the compressor.

Safety valve (-s) protect(s) compressor against internal overpressure. Frame reservoir's safety valve is adjusted at factory either to 9,5 bar (HK400) or 13,5 bar (HK 1000). (Additionally, the compressor block of HK1000 is equipped with two safety valves: the safety valve of 1st stage is adjusted at factory to 4 bar, whereas the 2nd stage valve is adjusted 15 bar).

Compressor is equipped with a built-in lubrication system.

PRESSURE SWITCH E (OPTIONAL):

Provides operation control in accordance with pre-adjusted maximum and minimum pressure limits in automatic mode.

When there is no air consumption compressor being on, pressure switch halts the unit de-energizing a hydraulic solenoid valve (i.e. by cutting off the hydraulic flow). Simultaneously the pressure in a hose between compressor block and frame reservoir is being let off making compressor's restart easy.

PNEUMATIC CUT-OFF PNE (OPTIONAL):

Provides operation control in accordance with pre-adjusted maximum and minimum pressure limits in automatic mode.

When there is no air consumption compressor being on, pneumatic cut-off halts the unit switching over the hydraulics to the free circulation. Simultaneously the pressure in a hose between compressor block and frame reservoir is being let off making compressor's restart easy.

PRESSURE AND COMPRESSION RATIO

The rated absolute intake pressure is 1 bar (100 000 Pa). (NOTE! The reference barometric pressure at 0 m above sea level is 1,0135 bar, when at 1000 m above sea level it drops to approx. 0,9 bar). When the discharge (operational) pressure of HKL-compressor is 8 bar, compression ratio is equal to:

$$\frac{\text{operational pressure} + \text{barometric pressure}}{\text{barometric pressure}} = \frac{8 + 1,0135}{1,0135} = 8,89$$

The compression ratio for unit of 12 bar is calculated in the same way.

AIR AND OIL TEMPERATURES

Filled with specified compressor oil or corresponding, (ref. to "COMPRESSOR OILS") unit is operated without any particular arrangements within ambient temperature range -5 °C... + 40 °C. If ambient temperature drops below -5 °C or raises over + 45 °C, fill the compressor with synthetic oil or be kindly requested to contact DYNASET's representative.

According to the standard EN 1012-1 the maximum accepted temperature of oil is +100 °C. In normal operational conditions at ambient temperature of +20 °C compressor oil's temperature levels to 80 - 85 °C.

The temperature of the delivered compressed air exceeds ambient temperature by approx. 5 - 10 K.

AIR HUMIDITY

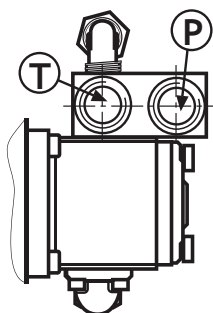
Compressor can be operated at the relative humidity up to 90%, provided that the ambient temperature is within the above range. If air humidity exceeds 90%, please contact DYNASET's representative.

INSTALLATION

DYNASET hydraulic compressor can be placed deliberately ensuring free air intake, sufficient cooling and an easy access to the unit. Position of a compressor should be horizontal. To protect compressor against overheating, do not allow the warmed cooling air to be directed by cooling fan back to the compressor block. Compressor's location should be chosen as far as possible from any source of dust and heat. Protective cover against weather and season impacts is recommended.

Intake air should not contain any smoke, toxic or flammable gas/vapor. When necessary, intake pipe can be fitted (the pipe should be measured properly in order to avoid load loss).

DYNASET hydraulic compressor works installed into an original hydraulic system of any carrier machine. Additionally, return line must be connected to a hydraulic oil tank directly in the shortest possible way in order to keep return oil pressure under 5 bar. Generally DYNASET's T-line is to be connected directly to the return line of a hydraulic system, when hydraulic fluid flows to the tank only through the return filter and cooler. Both filtering degree and cooling capacity of a hydraulic system should be sufficient (ref. to PARAMETERS).



It is possible to use DYNASET either as a mobile compressor and connect hydraulic power with quick couplings, or make a fixed installation on a carrier machine.

To put DYNASET compressor into operation only pressure- (P) and return (T) lines of a hydraulic system are to be connected to the unit's corresponding hydraulic ports. The proper connection ensures correct rotation direction of the compressor and, by that, an appropriate cooling of the compressor block. Rotation in opposite direction is not allowed. Furthermore, cross-connection of hydraulic hoses damages hydraulic motor's shaft seal.

Ensure that the oil flow is sufficient to run the unit, i.e. at least the nominal flow must be available.

At the oil flow less than demanded nominal rate, compressor's delivery will be below the nominal. If the oil flow is so excessive, that the automatic speed control valve is unable to handle it, the unit could be damaged. In the latter case the oil flow is to be reduced either by dropping down the rotation speed of hydraulic pump or using flow limiter valve. **DYNASET priority valve is recommended.**

After having been connected to the hydraulic system, DYNASET hydraulic compressor is to be started by directing the oil flow to the unit's pressure port by means of control valve - manual or solenoid. Even simple ball valve or three-way valve can be used.

To ensure DYNASET's proper operation, the hydraulic pressure must be set to the level, at which at least the nominal power output is achieved (ref. to PARAMETERS). Lower hydraulic pressure reduces compressed air pressure. Maximum hydraulic pressure is 250 bar (ref. to PARAMETERS). If the hydraulic system main pressure is higher, service pressure relief valve must be used.

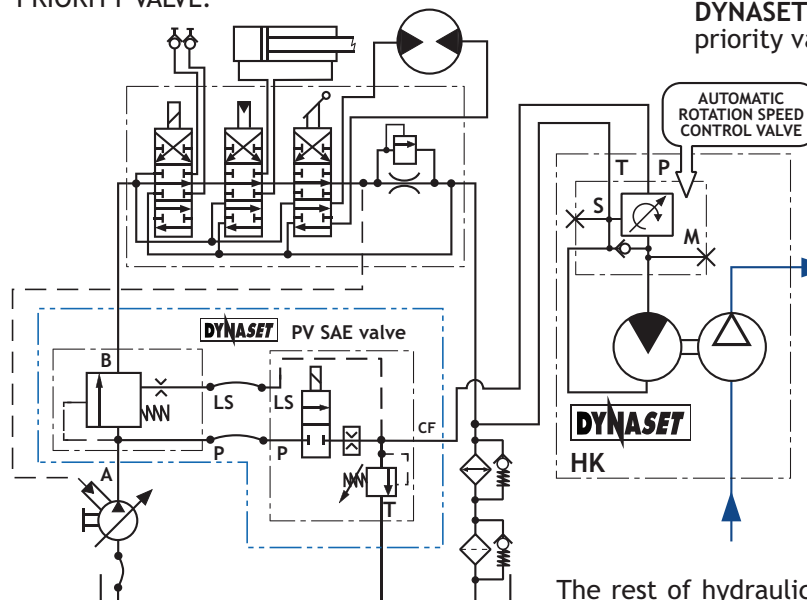
Ensure also that your machine oil cooling is working properly when DYNASET is in operation.

... INSTALLATION

DYNASET hydraulic compressors are designed for easy integration into hydraulic system of any type.

1. OPEN CENTRE HYDRAULIC SYSTEM WITH VARIABLE DISPLACEMENT PUMP

The demanded hydraulic flow is to be ensured and controlled with a DYNASET PRESSURE COMPENSATED PRIORITY VALVE.



DYNASET PV SAE 3/4 - 1 1/4 -XX lpm - 12/24 V priority valve includes following components:

1. Sandwich-mounted pressure compensator with SAE-flange specification;
2. Solenoid valve 12/24V;
3. Flow limiter;
4. Pressure relief valve.

PV SAE priority valve is designed for the installation to main pressure line between SAE-flanges of main hydraulic pump. Pre-adjusted, independent from other functions and prioritised hydraulic flow for DYNASET-unit comes from the solenoid valve.

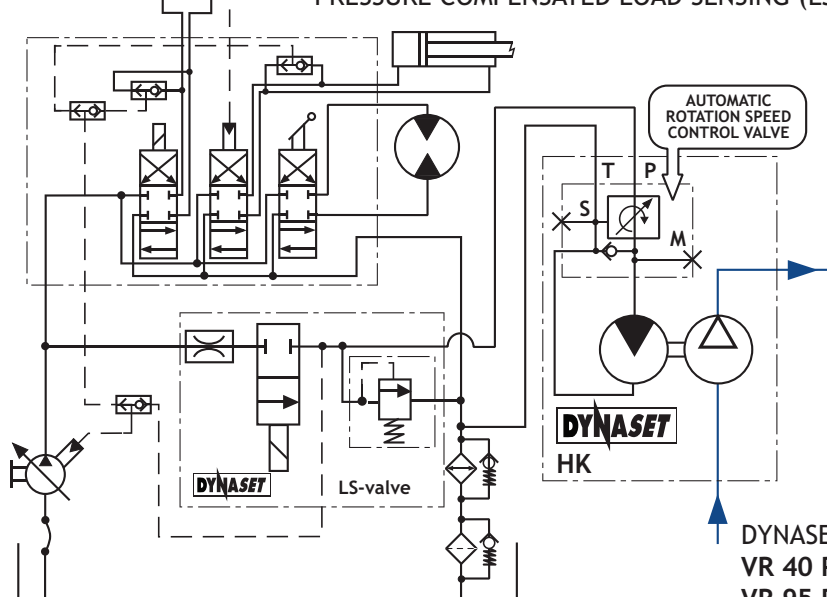
The rest of hydraulic pump's capacity is available for all other functions. Furthermore, pump's control works together with DYNASET PV SAE priority valve.

ALSO AVAILABLE for open centre hydraulic systems with variable displacement pump: PRIORITY VALVE PV C3C 3/4 - 1 1/4 - 12/24 V.

DYNASET PRIORITY VALVE enables trouble-free operation of your DYNASET-unit simultaneously with other hydraulic executors.

2. CLOSED CENTRE HYDRAULIC SYSTEM WITH VARIABLE DISPLACEMENT PUMP

The demanded hydraulic flow is to be ensured and controlled with a DYNASET PRESSURE COMPENSATED LOAD SENSING (LS) VALVE.



DYNASET LS-valve includes following components:

1. Flow limiter;
2. Solenoid valve 12/24V;
3. Pressure relief valve;
4. Shuttle valve (OPTIONAL).

DYNASET LOAD SENSING VALVES:

LSV 40, LSV 60, LSV 95 and LSV 150

LS-connection, pressure relief, max. hydraulic flow 40, 60, 95 ja 150 l/min respectively.

DYNASET FLOW LIMITERS to the pressure line:

VR 40 PK - 1/2, max. 35 l/min, with pressure compensation;

VR 95 PK - 3/4, max. 95 l/min, with pressure compensation.

VRD 180 PK - 1, max. 180 l/min, with pressure compensation.

ALSO AVAILABLE for closed centre hydraulic systems with variable displacement pump:

DYNASET SOLENOID VALVE to the pressure line for remote starting.

SV 70 NC - 1/2 - 12 / 24 V max 70 l/min with LS -connection;

SV 150 NC- 1 - 12 / 24 V max 150 l/min with LS -connection.

OPERATION

Prior to starting the compressor check whether the unit is filled with oil. In non-operating and depressurized compressor oil level is at the midlevel of sight glass. When necessary add oil to the sight glass midlevel. Add oil of the same quality as the unit was filled with. Replace and tighten the filler plug.

Open the cut-off valve in compressor's pressure line. Ensure from compressor's pressure gauge that the unit is decompressed - when pressurized, compressor may start with difficulty*).

With carrier's engine running HK-compressor is started with a hydraulic control valve. Compressor starts to produce usable compressed air when hydraulic flow achieves the functional minimum rate, which is approx. a half of the nominal value (Ref. to the unit's nameplate as well as to the PARAMETERS). The nominal discharge is achieved when the compressor is rotated by hydraulic flow adequate to the nominal rotation speed.

*)

When the compressor is to be started under pressure (for instance, air reservoir in pressure line) or the unit is used intermittently stopping and restarting it several times according to the air demand, following accessories are recommended:

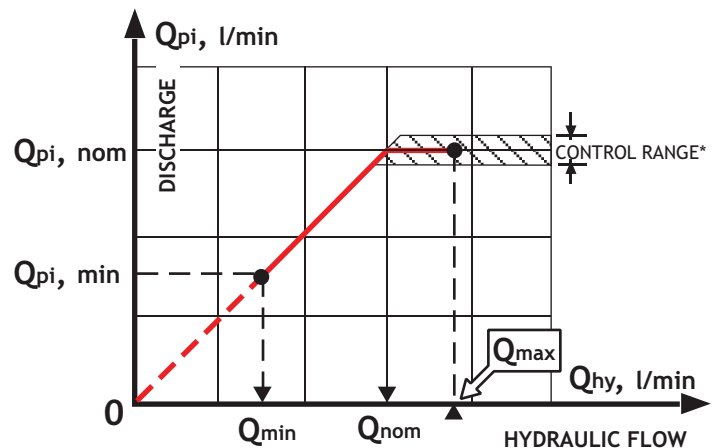
- * Electric pressure switch (-E);
- * Pneumatic cut-off (-PNE).

Compressor being rotated at full output, automatic rotation speed control valve keeps rotation speed constant when incoming hydraulic oil flow (Q_{hy}) can vary from nominal demanded flow (Q_{nom}) up to value exceeding Q_{nom} by 10 - 20 l/min depending on compressor size.

$$Q_{hy_nom} = Q_{hy_calc} + 2 \text{ l/min}$$

Q_{hy_calc} - CALCULATED CONSTANT FLOW
 Q_{hy_nom} - NOMINAL FLOW

$$Q_{hy_min} \approx 0,5 Q_{hy_nom}$$



OFF-LOAD MODE (IN STANDARD ASSEMBLY)

Compressed air is consumed according to the demand, when compressor's delivery levels within 0 l/min to the maximum. If the compressed air is not used while compressor is running (i.e. delivery line is cut off), unit's internal pressure achieves the maximum when the unloader valve opens and lets off the air, compressor's output as well as pressure in frame reservoir being maximum.

Unloader's built-in check valve keeps the frame reservoir pressurized.

When the compressed air is requested (i.e. delivery line is open), pressure in frame reservoir decreases and, consequently, unloader valve closes and re-directs the air into delivery line.

ATTENTION !

The compressor must not be operated neither in smoky environment nor in conditions where toxic or flammable vapors could be aspirated.

Compressor's maximum pressure must not be exceeded in any circumstances.

Compressed air delivered by a HKL-compressor must not be used as the breathing air, even if purified and filtered. HKL-compressor is designed for air only and must not be used with any other gas.

If the compressor is connected to the distribution network or consumer with a flexible hose, ensure that the hose is of proper diameter and pressure class. Damaged or worn hose should be replaced immediately.

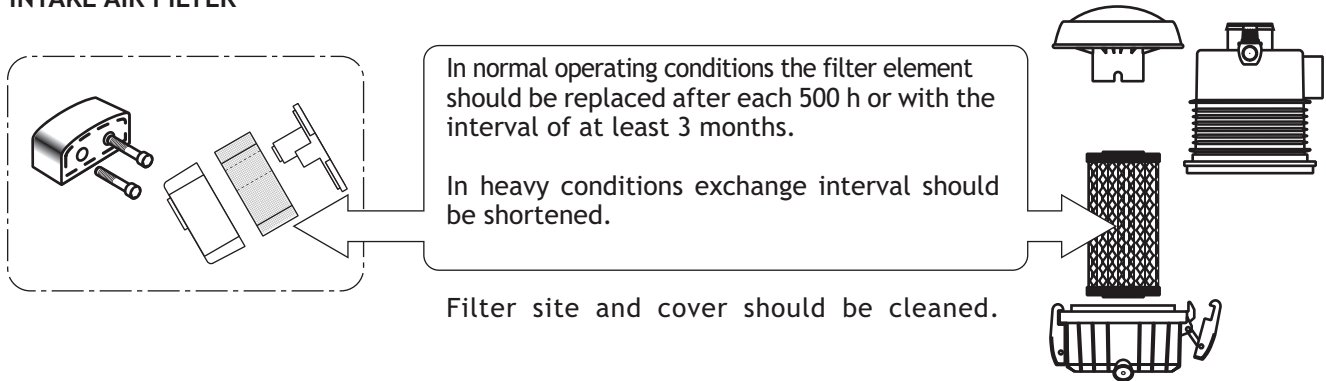
The sound-proof enclosure (if used) should be closed when compressor is in the operation.

MAINTENANCE

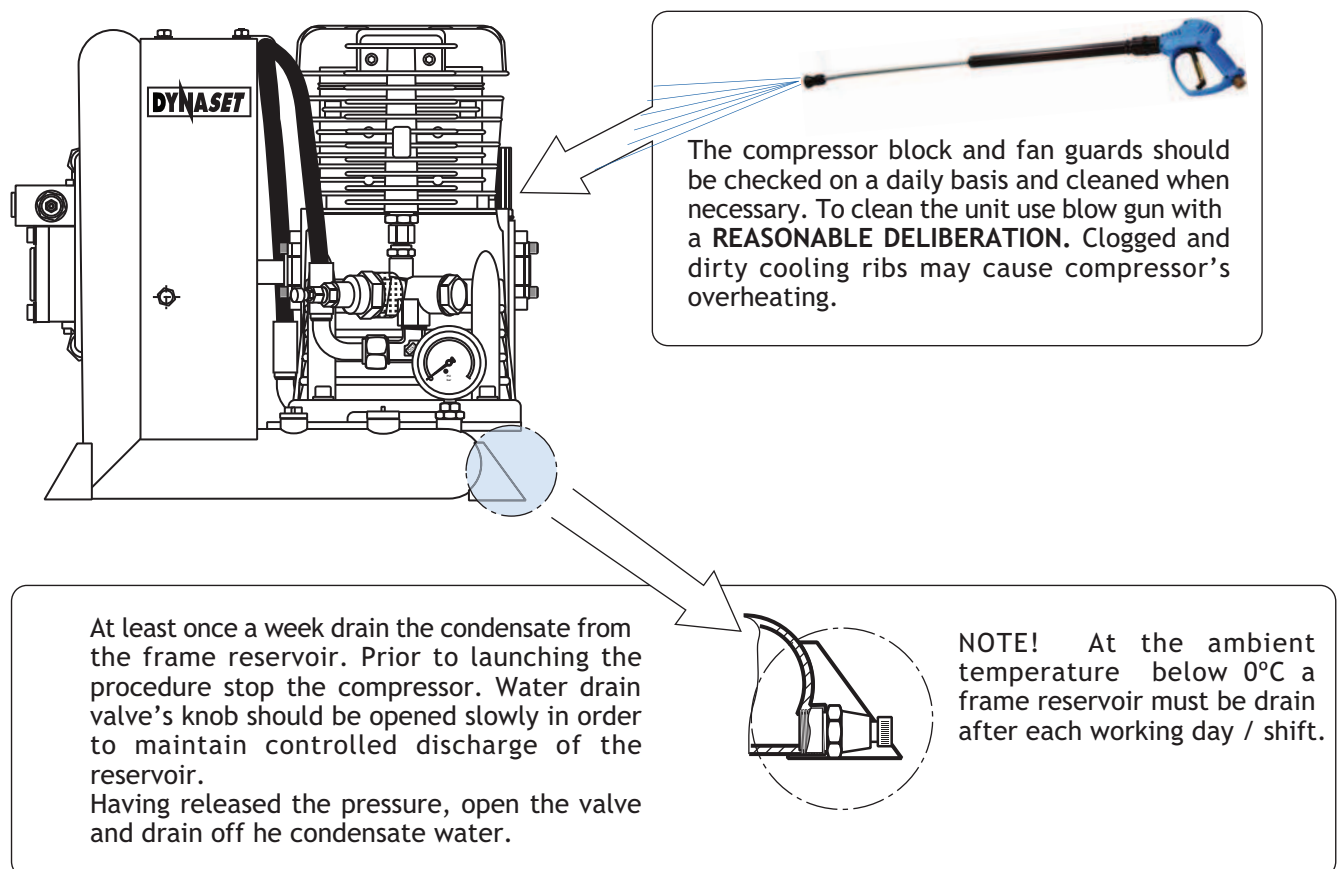
DYNASET hydraulic compressors are low-maintenance units. Only normally wearing parts and materials should be replaced either when necessary or in accordance with a service program. Refer also to the TROUBLESHOOTING section.

Prior to starting any maintenance or repair, ensure that the compressor is completely depressurized. The unit's pressure gauge must indicate zero.

INTAKE AIR FILTER

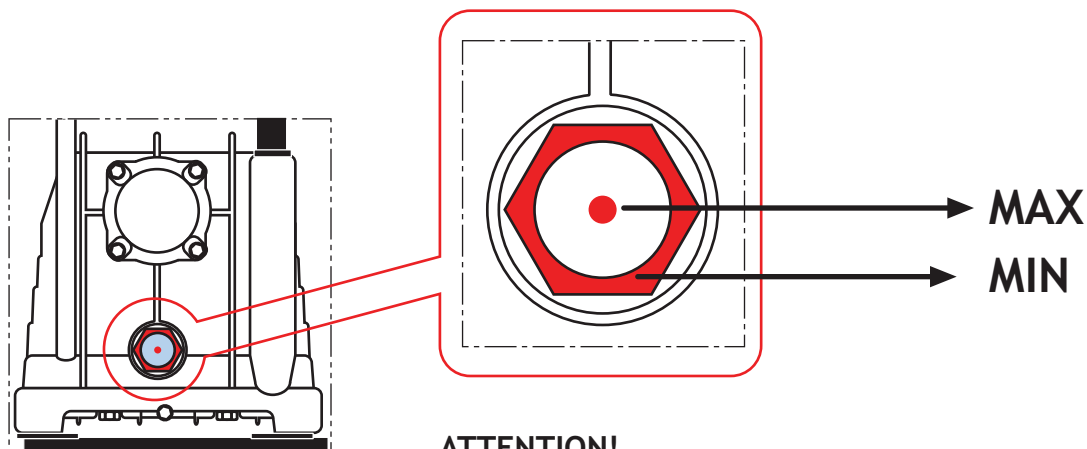


COMPRESSOR BLOCK MAINTENANCE



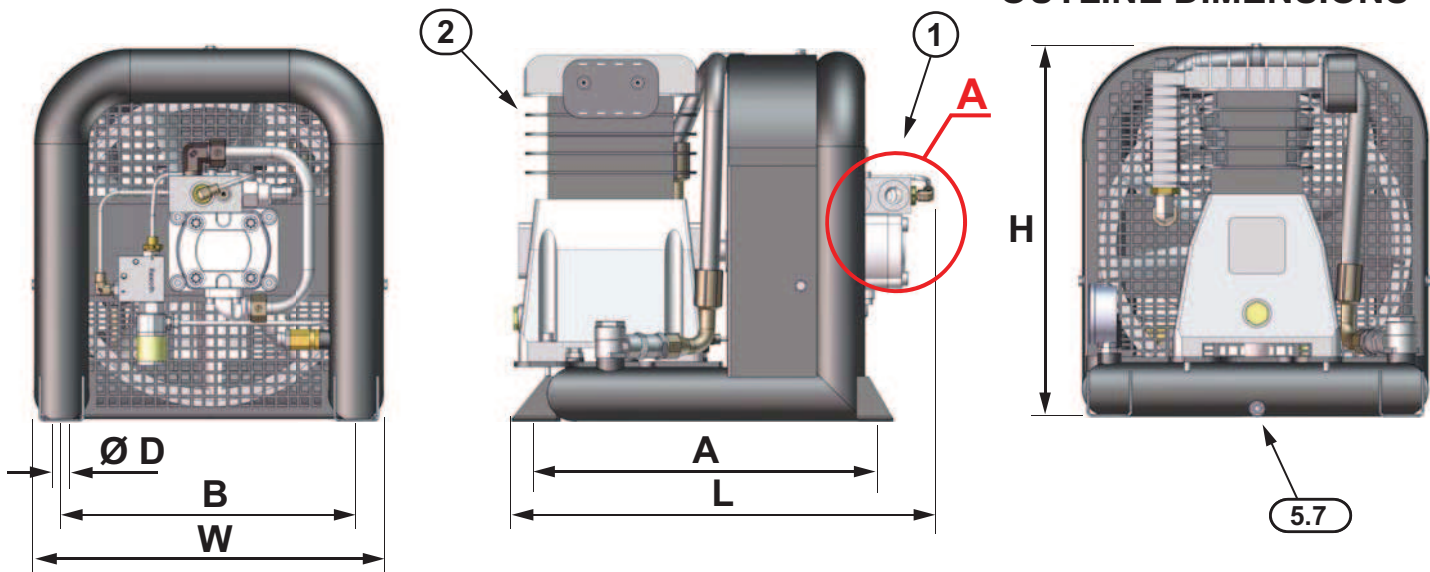
THE FOLLIWING REPAIR KITS TO THE COMPRESSOR BLOCK ARE AVAILABLE:

- 1 SEAL KIT;
- 2 PISTON REPAIR KIR, incl. pistons, oil rings, piston pins and seal rings;
- 3 CRANKSHAFT BEARING KIT.

OIL LEVEL CHECK**ATTENTION!**

OIL LEVEL IN A RUNNING COMPRESSOR UNIT MUST BE BETWEEN THE BOTTOM OF THE SIGHTGLASS (MIN) AND RED POINT (MAX) IN THE CENTRE OF IT.

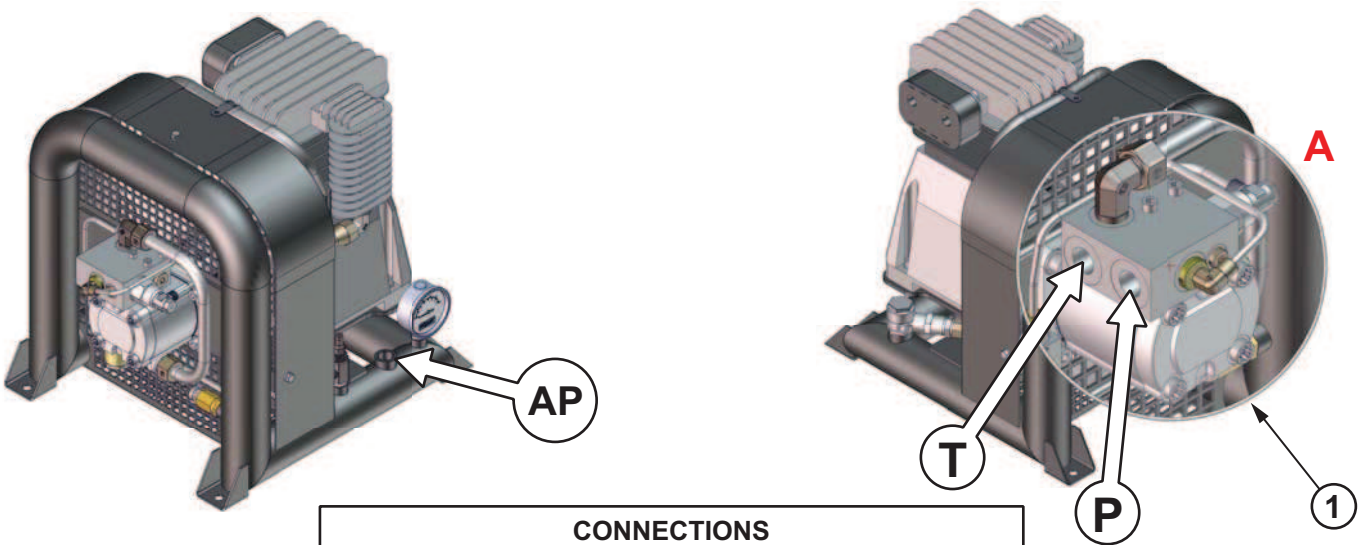
OUTLINE DIMENSIONS



MODEL	DIMENSIONS, mm							WEIGHT	OIL FILL
	L	W	H	A	B	C	D	kg	l
HK 400/8 - 23	475	400	415	395	325	-	11	37	0,9

HYDRAULIC FLOW	l/min, max.	40
HYDRAULIC PRESSURE	bar, max.	250
AIR FLOW	l/min, max.	400
AIR PRESSURE	bar, max	8
WEIGHT	kg	37

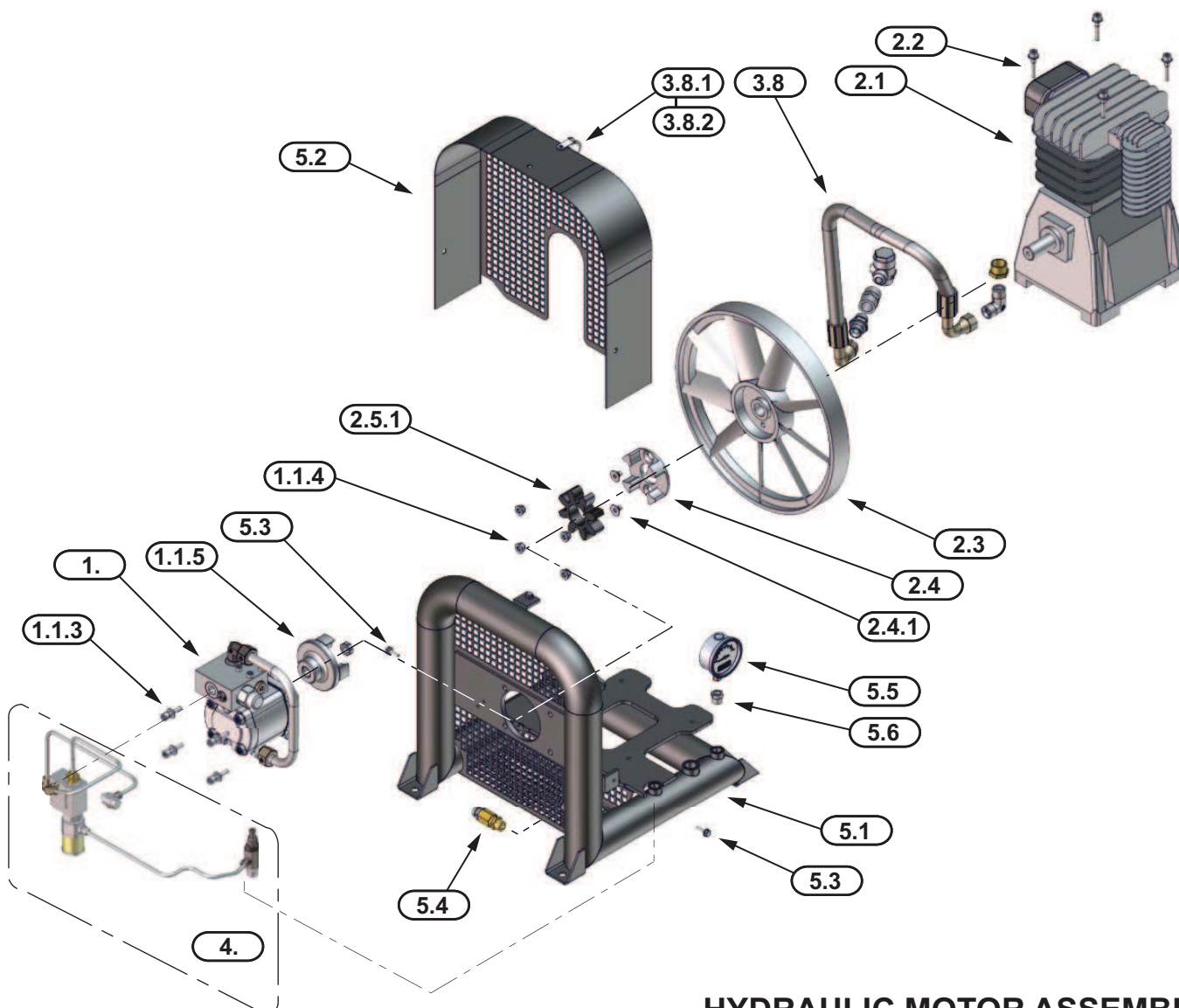
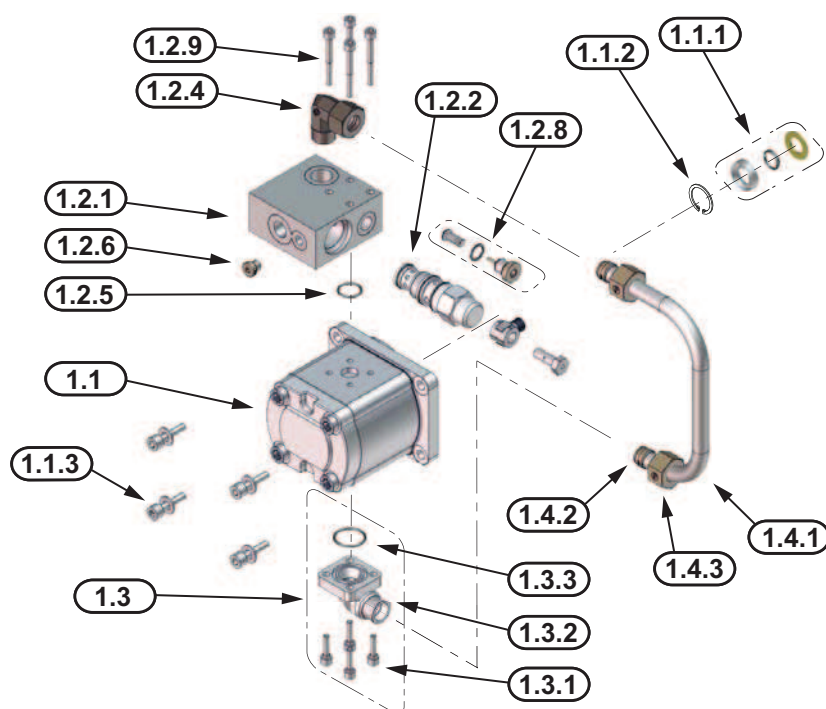
HYDRAULIC AND COMPRESSED AIR CONNECTIONS

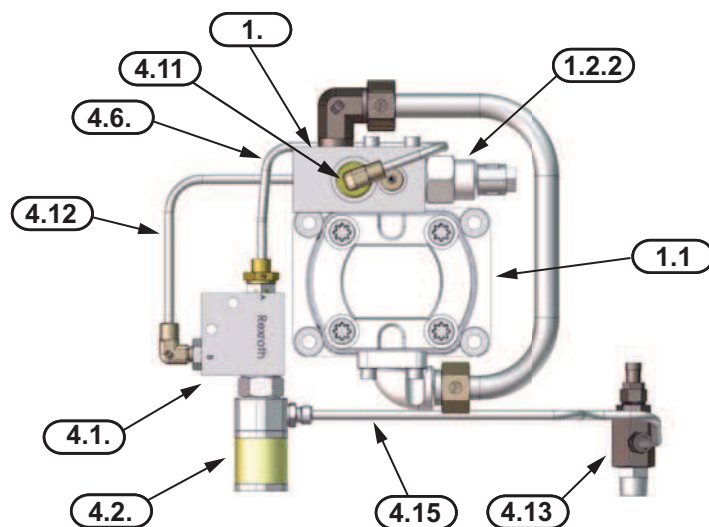
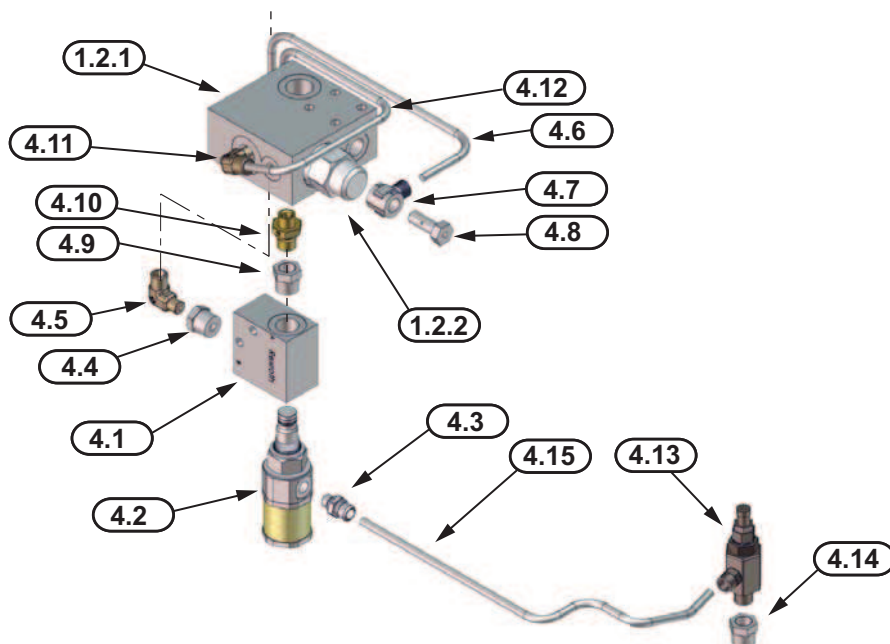
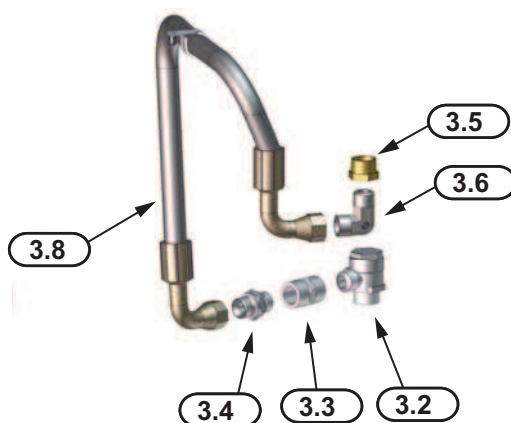


CONNECTIONS			
HYDRAULICS			COMPRESSED AIR
P pressure	T return	D drain	AP
BSP 1/2"	BSP 1/2"	-	BSP 1/2"

PARAMETERS

PARAMETERS		HK 400 /8 - 23_PNE
DISCHARGE CHARACTERISTICS		
Flow rate at ref. conditions, max. l/min		400 *
Maximum pressure	bar	8
COMPRESSED AIR CONNECTION	AP	BSP 1/2"
HYDRAULIC CONNECTIONS		
Pressure line	P	BSP 1/2"
Return line	T	BSP 1/2"
FRAME RESERVOIR	I	4,4
HYDRAULIC POWER REQUIREMENTS		
Minimum flow	l/min	12
Maximum flow	l/min	40
Pressure at nominal flow	bar	210
Maximum pressure	bar	250
Pressure when unloaded	bar	50
HYDRAULIC FLUID REQUIREMENTS		
Viscosity	cSt	10 - 200 / optimi 25 - 35
Temperature	°C	max. 70 **
Filter ratio, recommendation	um	min. 25
Cooling capacity, recommend. min	kW	2
OVERALL DIMENSIONS		
Length	mm	455
Width	mm	390
Height	mm	415
Weight	kg	37
* According to ISO 1217 (1996) at discharge pressure of 6 bar.		
** Ref. to HYDRAULIC FLUIDS		

MAIN ASSEMBLY**HYDRAULIC MOTOR ASSEMBLY**

HYDRAULIC MOTOR + PNE ASSEMBLY**PNE ASSEMBLY****COMPRESSOR BLOCK TUBE ASSEMBLY**

PART LIST

NO	CODE	DESIGNATION	DETAILS	HK 400/8-23_PNE
1	D100104270	HYDRAULIC MOTOR ASSEMBLY		
1.1	0700110060	HYDRAULIC MOTOR	16R	1
1.1.1	V100103160	SHAFT SEAL, COMPLETE	17x30	1
1.1.2	V100103300	SNAP RING	30	1
1.1.3	8500412300	ALLEN SCREW	M8x30	4
1.1.4	8500905800	NUT NYLOC	M8	4
1.1.5	8107196000	HALF COUPLING, MOTOR SIDE	H86-2r	1
1.2	0709011181	RPM-VALVE BLOCK ASSEMBLY	3007D	1
1.2.1	0700031435	RPM-VALVE BLOCK		1
1.2.2	0700001144	RPM-CARTRIDGE	FLC-1	1
1.2.4	1702015000	ELBOW FITTING	15L RK1/2 DIN	1
1.2.5	2204104000	O-RING 15		1
1.2.6	1702690000	PLUG	HF R1/8	1
1.2.8	1309010276	CHECK VALVE COMPLETE	R1/4	1
1.2.9	8500410000	ALLEN SCREW	M6x50	4
1.3		RETURN LINE FLANGE COMPLETE	15L RK1/2 DIN	1
1.3.1	8500409400	ALLEN SCREW	M6x20	4
1.3.2	1100101000	SUCTION FLANGE FITTING	15	1
1.3.3	2204134000	O-RING 24		
1.4	0700001445	RETURN PIPE ASSEMBLY, COMPLETE	D15x171x45	1
1.4.1	8707778000	PIPE	15x1,5 ST37,4	1
1.4.2	1701856000	FERRULE	15	2
1.4.3	1701870000	NUT 15L	M22x1,5	2
2		COMPRESSOR BLOCK, ASSEMBLY		1
2.1	7106811000	COMPRESSOR BLOCK	HK400	1
2.2	8500412500	HEX SCREW	M8x40	4
	8501126000	WASHER	8,4x16	4
2.3	0600040576	BLOWER WHEEL	HK400	1
2.4	0600040575	HALF COUPLING COMPRESSOR SIDE	86	1
2.4.1	8500584230	ALLEN SCREW	M10x12	2
2.5.1	8107214000	RUBBER ELEMENT	H86	1
3.2	7306953000	CHECK VALVE	R3/4" x 1/2"	1
3.3	1702520000	ADAPTER	1/2"	1
3.4	17024xxxxx	ADAPTER	1/2"u 1/2"u	1
3.5	1702940000	FITTING	RK3/4u-1/2s	1
3.6	1702370000	ELBOW FITTING	R1/2xRK1/2	1
3.8		COMPR. BLOCK TUBE ASSEMBLY	R08-90/08-520-1-N/R08-90/45	1
3.8.1	8307652400	HOSE CLAMP	22	1
3.8.2	8500006000	HEX SCREW	M6X20	2
4.		PNE ASSEMBLY		1
4.1	1300551000	BODY CARTRIDGE	2/2 3/8"	1
4.2	1300545000	VALVE CARTRIDGE	2/2 NC 25lpm	1
4.3	1701880000	ELBOW FITTING	06L R1/8	1
4.4	1702920000	REDUCER	RK3/8u-1/8s	1
4.5	1701940000	ELBOW FITTING	06L RK1/8	1
4.6		TUBE 6X1		m
4.7		BANJO FITTING	6L	1
4.8		BANJO SCREW	UNF 3/8-24	1
4.9	1702925000	REDUCER	RK3/8u-1/4s	1
4.10	1701885000	FITTING	06L RK1/4	1
4.11	1701940000	ELBOW FITTING	06L RK1/8	1
4.12		TUBE 6X1		m
4.13	7306986000	PILOT VALVE	R1/4", 4-12bar	1
4.14	1702925000	REDUCER	RK3/8u-1/4s	1
4.15		TUBE 6X1		m
5.		COMPRESSOR FRAMEWORK		1
5.1	0600041598	AIR RESERVOIR	4.4L	1
5.2	0600041618	BLOWER GUARD	HK400	1
5.3	8500006000	HEX SCREW	M6X20	2
5.4	7306971500	SAFETY VALVE	3/8s	1
5.5	1501304000	PRESSURE GAUGE, DAMPENED	RK1/4	1
5.6	1702925000	REDUCER	RK3/8u-1/4s	1
5.7	7106945000	WATER DRAIN VALVE	1/4"u	1

NOTE !

WHEN ORDERING SPARE PARTS, UNIT'S TYPE, MODEL AND SERIAL NUMBER SHOULD BE INDICATED.

HYDRAULIC FLUIDS

Wide range of standard hydraulic fluids can be used with the DYNASET hydraulic equipment. Depending on the operating temperature, following mineral hydraulic oils are recommended:

ISO VG 32S	for oil's operation temperature up to 70 °C;
ISO VG 46S	for oil's operation temperature up to 80 °C;
ISO VG 68S	for oil's operation temperature up to 90 °C.

Synthetic and bio-oils can be used as well if their viscosity characteristics and lubricating efficiency are corresponding to above mineral oils. Automatic transmission fluids and even engine oils can be used, provided that they are allowed to be used in hydraulic system of your carrier machine.

To use special hydraulic fluids a with DYNASET equipment, please be kindly requested to contact nearest DYNASET representative for an advice.

LUBRICATION OILS

QUALITY	AMBIENT TEMPERATURE	CHANGE INTERVAL, MAXIMUM	OIL FILL	
			HK400	HK1000
SAE 10W-30	- 5 °C ... + 40 °C	500 h */ 12 months	0,91 l	3,1 l
ATTENTION ! FACTORY-FILL OIL MUST BE CHANGED AFTER VERY FIRST 150 (MAX.) OPERATION HOURS !				

DIESEL ENGINE OILS		PARAMETERS				AMBIENT TEMPERATURE		
STANDARD	CHANGE INTERVAL, MAXIMUM					-25...+20°C	-5...+30°C	+5...+40°C
API CC	500 h/ 3 months	SAE-class				10W	30	40
MIL-L-2104		Viscosity + 40°C	cSt			35 - 45	100 - 110	145 - 160
MIL-L-46152 B		Viscosity + 100°C	cSt			5,6 - 9,3	9,3 - 12,5	12,5 - 16,3
		Pour point	°C			- 30	-20	-15
		Flash point	°C			210 - 220	230 - 240	240 - 250
		Specific weight	kg/m³			880	890	900

TROUBLESHOOTING

1. COMPRESSOR DOES NOT START WHILE CARRIER'S ENGINE IS RUNNING AND HYDRAULIC SYSTEM IS ON.	1.1 Malfunction of compressor's hydraulic control valve. 1.2 Compressor's excessive internal pressure. 1.3 Failure in hydraulic system. 1.4 Air pressure in delivery line interferes compressor's start.	1.1 Check and repair. 1.2 Release the frame tank's pressure, for instance, by means of 3-way valve installed to the delivery line). In case of intermittent / cyclic duty, either electric pressure switch (-E) or pneumatic cut-off (-PNE) is recommended. 1.3.1 Check whether the hydraulic fluid flow and pressure are sufficient. Adjust when necessary. 1.3.2 Check the hydraulic motor for possible leakage. Replace motor if necessary. 1.4 Let off the pressure from the delivery line prior to starting the compressor. Ref. to 1.2
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2. COMPRESSOR STARTS HARDLY	See 1.3 See 1.4	
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3. AIR PRESSURE DOES NOT ACHIEVE REQUESTED RATE.	3.1 Safety valve's failure or incorrect setting. 3.2 Failure in hydraulic system. 3.3 Intake air filter is clogged.	3.1 Check and adjust / replace when necessary. 3.2.1 Check whether the hydraulic fluid flow and pressure are sufficient. Adjust when necessary. 3.2.2 Check the hydraulic motor for possible leakage. Replace motor if necessary. 3.3 Check and replace.
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... TROUBLESHOOTING

4. COMPRESSOR DOES NOT PRODUCE COMPRESSED AIR = RUNS UNLOADED	4.1 Delivery line closed (i.e. OPEN/CLOSE valve is damaged) or possible malfunction in pneumatic executor. 4.2 Unloader valve's failure or incorrect setting. 4.3 Clogged intake air filter /air intake port.	4.1 Check and repair possible failure. 4.2. Check and repair possible failure. 4.3 Check and clean port /replace filter element .
5. EXCESSIVE OIL CONSUMPTION.	5.1 Worn piston rings. 5.2 Compressor block filled with an improper oil.	5.1 Check and repair. 5.2. Fill the compressor block with an approved oil according to the season.
6. COMPRESSOR'S OVERHEATING.	6.1 Dirty compressor block, clogged cooling fan's guard. 6.2 High ambient temperature. 6.3 Low oil level. 6.4 Due to the insufficient cooling capacity of a hydraulic system, hydraulic fluid warms the compressor.	6.1 Check and clean. 6.2 Sufficient ventilation is to be arranged. 6.3 Add compressor oil to the required level. 6.4 DYNASET OY offers wide range of hydraulic components including oil coolers !

PRODUCT DISPOSAL

Conform to waste legislation, regulations and recommendations in waste disposal and waste recycling issued by local authorities.

Precondition:

- Product is put properly out of order.
- Before transportation get off all used agent (oil, cooling liquid) and dirty filters.

Items requiring specialist treatment:

- Separate the base materials, iron, copper, steel, electronic pcbs, removing paint, polyester resin, and insulation tape and/or plastics residues from all components.

Dispose of this 'waste material' can now be recycled.

Forward the reclaimed materials for recycling.

Waste material:

- Dispose of waste material from both of the above processes via an authorized disposal company.

Customer can send us the old Dynaset equipment for reuse or recycling at the purchaser's cost, adequately packed for shipment, to the DYNASET OY or to other location authored by DYNASET OY. Shipment documents must contain purchaser's name and contact information.

MANUFACTURER'S LIMITED WARRANTY

1. Warranty coverage

All hydraulic accessories manufactured by DYNASET OY are subject to the terms and conditions of this limited warranty. Products are warranted to the original purchaser to be free from defects in materials or workmanship. Exclusions from warranty are explained in item 8.

2. Beginning of warranty period

Warranty period begins from the delivery date of the product. Delivery is considered to be done on the date when installation has been accomplished or purchaser has taken the product in use. Product is considered as taken in use at the date when DYNASET OY has delivered the product to purchaser, unless separately agreed otherwise by written agreement.

3. Warranty period

Warranty period is twelve (12) months based on maximum of 2000 hours usage during this time period. In cases where the system is provided complete with certain special components (e.g. drive unit), those components are considered as a subject to their manufacturer's warranty.

4. Warranty procedures

Immediately upon identifying a problem which purchaser believes to be a failure subject to the product's limited warranty, purchaser must contact primary to the seller of the product. Contact must be made as soon as possible, latest thirty (30) days after the problem was identified. Seller and/or manufacturer technical staff determines the nature of the problem primarily by phone or e-mail. Purchaser commits to give necessary information and to perform routine diagnostic procedures in order to determine the nature of the problem and necessary procedures.

5. Warranty repairs

If the product is found to be defective during the warranty period, DYNASET OY will, at its option, either repair the product, author it to be repaired at its authorized workshop or exchange the defective product. If the product must be repaired elsewhere than premises of DYNASET OY or authorized workshop, all costs excluded from this warranty (traveling and waiting hours, daily allowance, traveling expenses and uninstallation/reinstallation costs) will be charged from the purchaser.

If the problem is not covered by this limited warranty, DYNASET OY has the right to charge purchaser of troubleshooting and repairing.

6. Delivery terms of warranty repair

If the product is found possible to be defective under this limited warranty and it needs to be repaired, DYNASET OY gives Warranty Return Number (WRN). Items being returned must be shipped, at the purchaser's cost, adequately packed for shipment, to the DYNASET OY or to other location authored by DYNASET OY.

Shipment documents must contain:

- Purchaser's name and contact information
- Receipt of original purchase
- WRN code
- Problem description

7. Warranty of repaired product

Warranty period of the product repaired under this limited warranty continues to the end of original warranty period.

8. Exclusions from warranty

This warranty shall not apply to:

- a. Failures due to normal wear and tear, improper installation, misuse, abuse, negligence, purchaser selection of improper product to intended use, accident, improper filtration of hydraulic oil or intake water or lack of maintenance
- b. Cost of maintenance, adjustments, installation or startup
- c. Coating, hydraulic oil, quick couplings and interconnection hoses (internal or external to system assemblies)
- d. Products altered or modified in a manner not authorized by DYNASET OY in writing
- e. Products which have been repaired during warranty period by others than DYNASET OY or its authorized workshop
- f. Costs of any other damage or loss, whether direct, indirect, incidental, special or consequential, arising out of the use of, or the inability to use, the product
- g. Telephone or other communications expense
- h. Product that is used in exceptional conditions, considered to cause excessive wear and tear
- i. Faults caused by nature phenomenon's like flood, thunder, etc.

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EC DECLARATION OF CONFORMITY

We hereby declare that the below-reference product design, structure and our manufacture form fulfills relevant, fundamental safety and health requirements with the provisions in the Council Directive on mutual approximation of laws of the Member States on the safety of machines.

- Machine directive 2006/42/EC
- Directive 97/23/EC European standards Pressure equipment.
- LVD directive 2006/95/EC

Apply conformity standards:

- CEN EN ISO 4413: EN ISO 4413:2010
Hydraulic fluid power - General rules and safety requirements for systems and their components.

If device has changed by someone other than at the hands of manufacturer or his permission, this declaration is not valid.

PRODUCT: HYDRAULIC COMPRESSOR

TYPE NUMBER: HK 400/8-23-xxx-x
HK 1000/12-35-xxx-x

YLÖJÄRVI 07.05.2012

DYNASET Oy



Timo Nieminen
R&D Manager

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